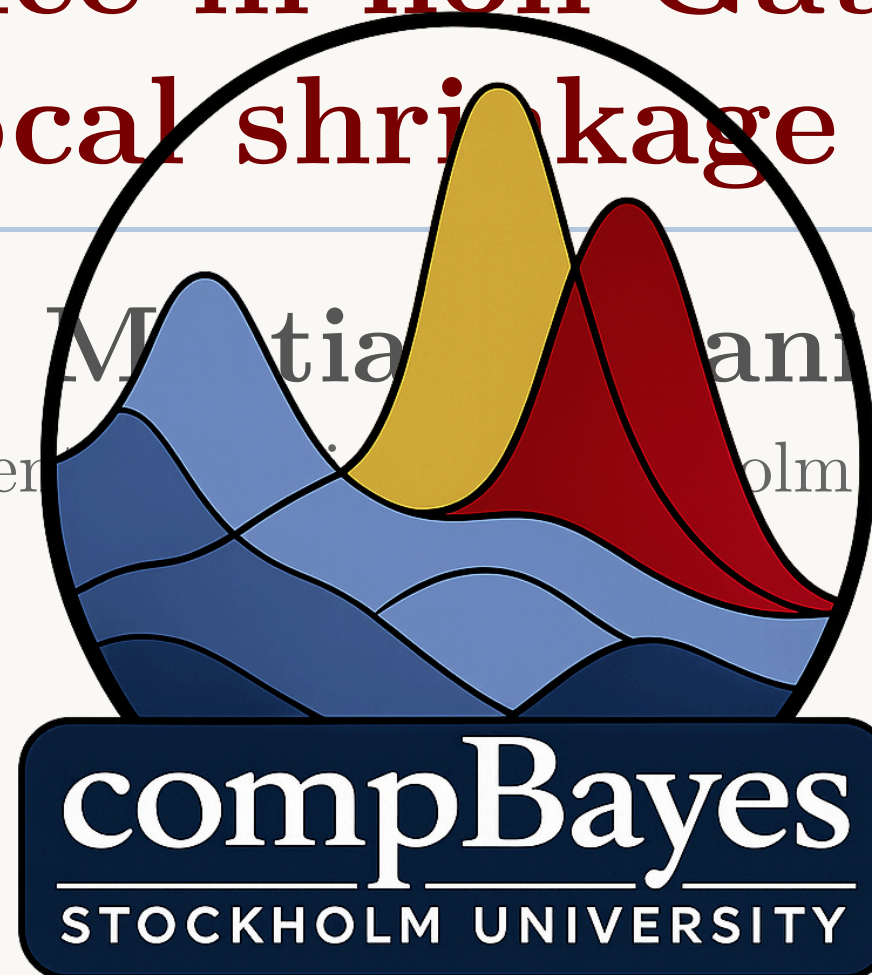


Bayesian inference in non-Gaussian time series with global-local shrinkage process priors

Department of Statistics, Stockholm University



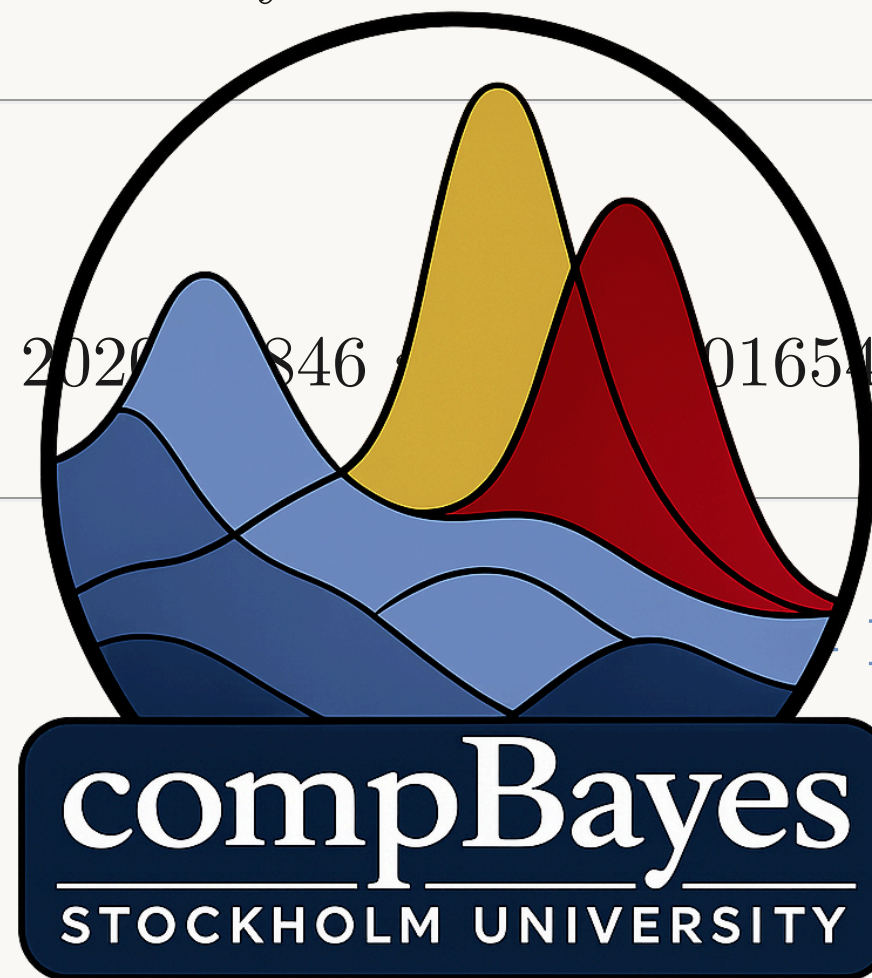
Collaborators

- **Robert Kohn**, University of New South Wales, Sydney.
- **Ganna Fagerberg**, Stockholm University
- **Yijie Niu**, Stockholm University
- **Oskar Gustafsson**, Stockholm University

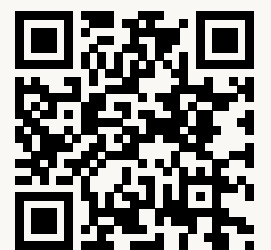
Funding

- Swedish Research Council, grants 2020-03846 and 2016-01654.

Slides

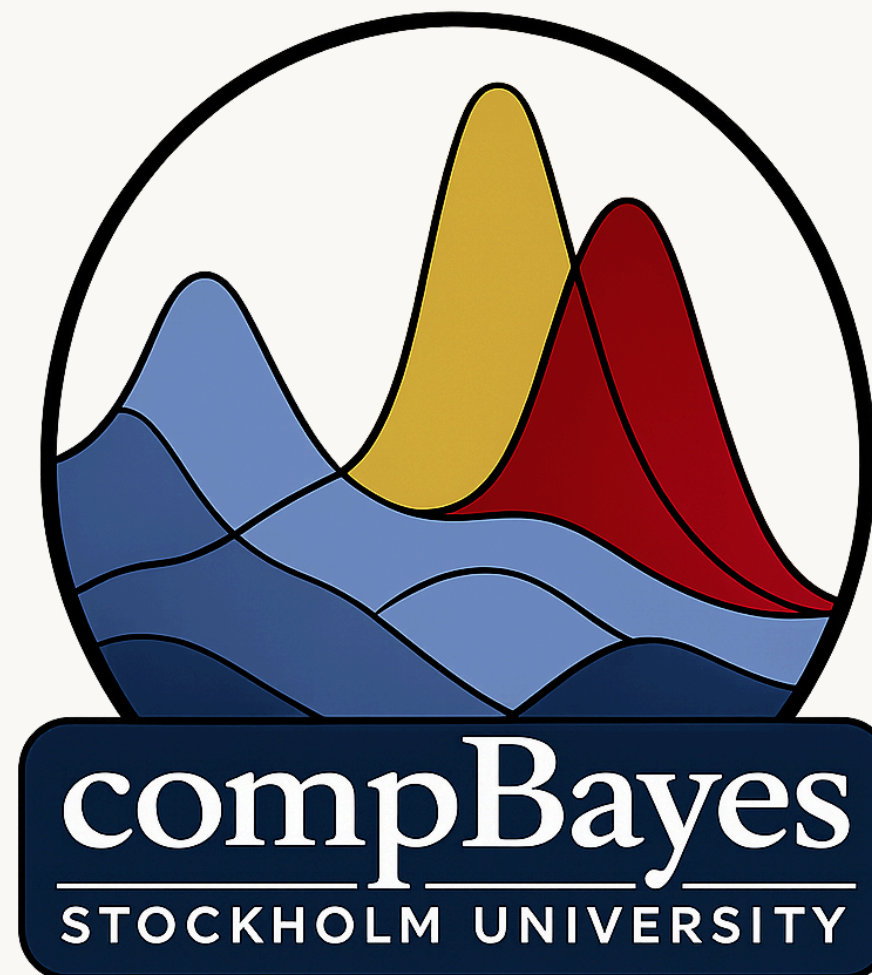


packages (unoptimized, ongoing)



Outline

0. Time-varying parameters and state-space models
1. Dynamic shrinkage process priors
2. Time-varying multi-seasonal ARMA
3. Non-Gaussian models
4. Posterior inference



Time-varying parameter models

- Regression with **time-varying parameters**

$$y_t = \mathbf{x}_t^\top \boldsymbol{\beta}_t + \varepsilon_t \quad \varepsilon_t \stackrel{\text{iid}}{\sim} N(0, \sigma_\varepsilon^2)$$

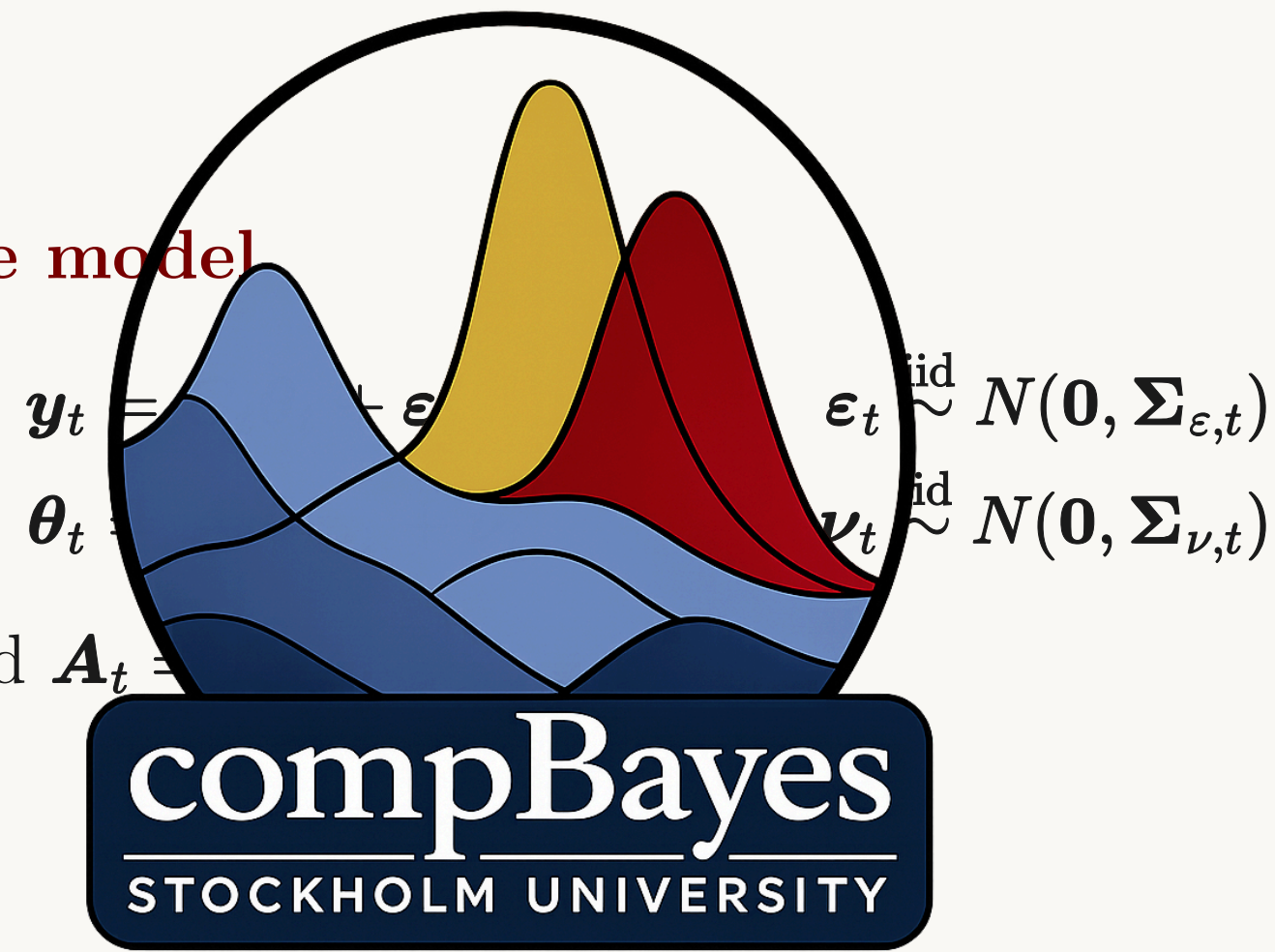
$$\boldsymbol{\beta}_t = \boldsymbol{\beta}_{t-1} + \boldsymbol{\nu}_t \quad \boldsymbol{\nu}_t \stackrel{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_\nu)$$

- Linear Gaussian state-space model**

Observation model:

Transition model:

with state $\boldsymbol{\theta}_t = \boldsymbol{\beta}_t$, $\mathbf{C}_t = \mathbf{x}_t^\top$ and $\mathbf{A}_t =$



Regression with time-varying parameters

time series length

100

σ_ε

1

σ_{β_0}

1

σ_{β_1}

1

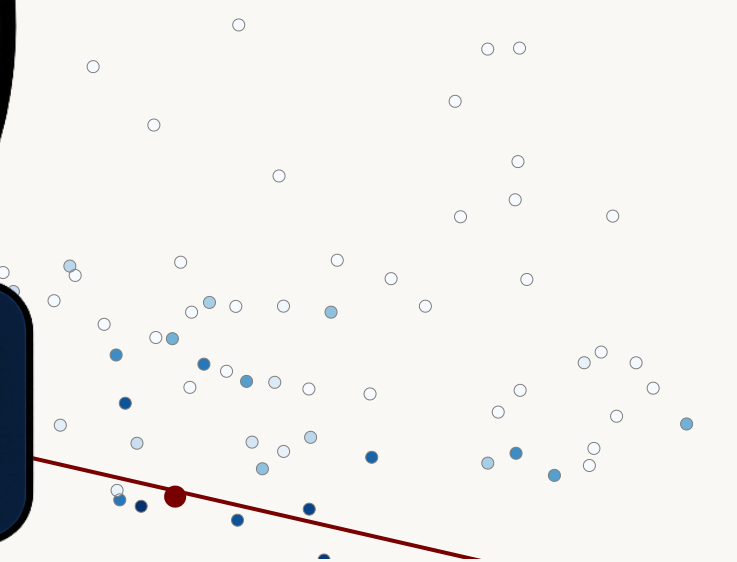
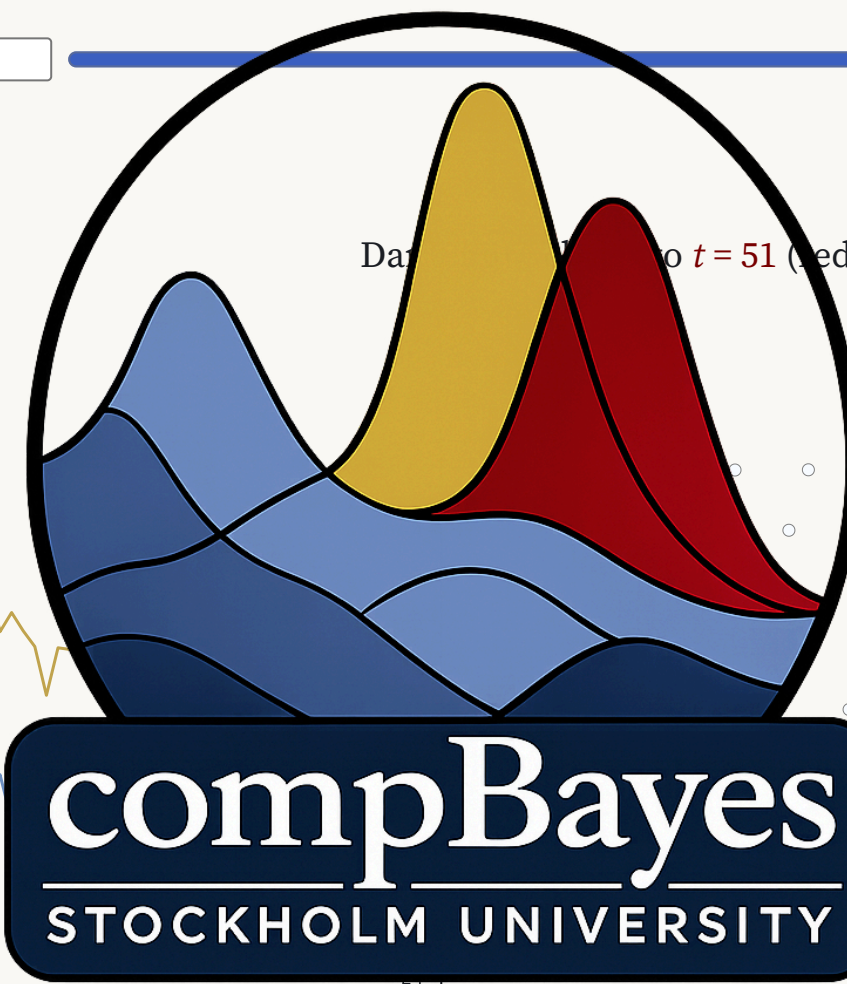
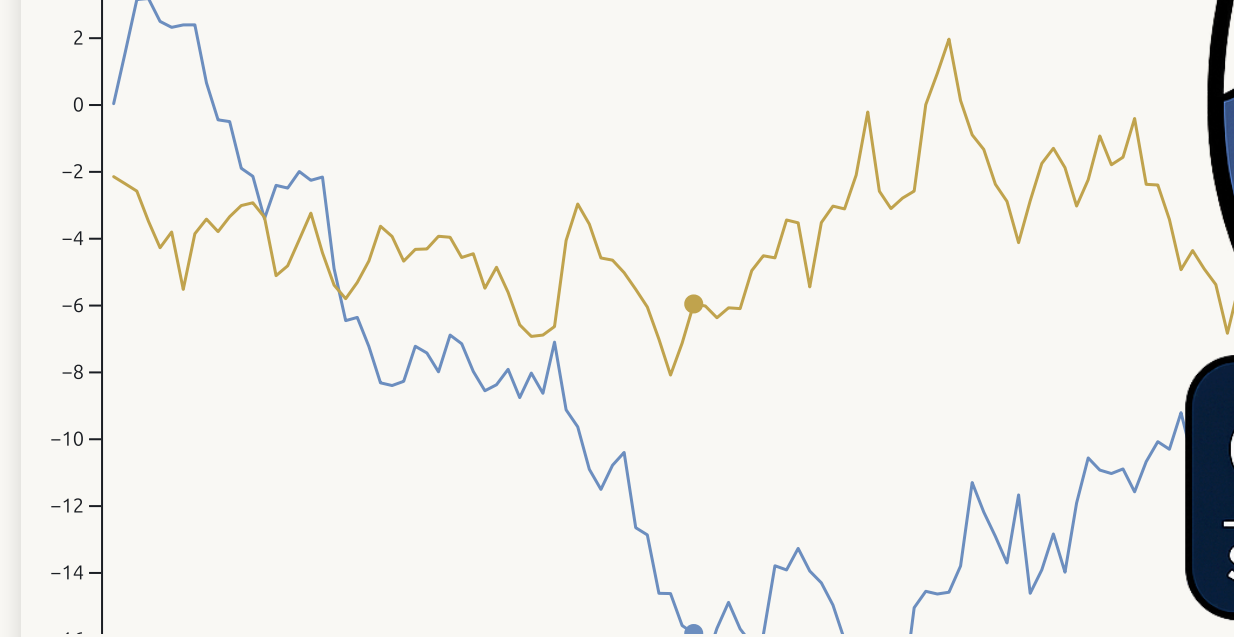
regr line at time

51

Simulate new data

Evolution: $\beta_0 = -15.812$ and $\beta_1 = -5.952$

coeffs



Taxonomy of state-space models

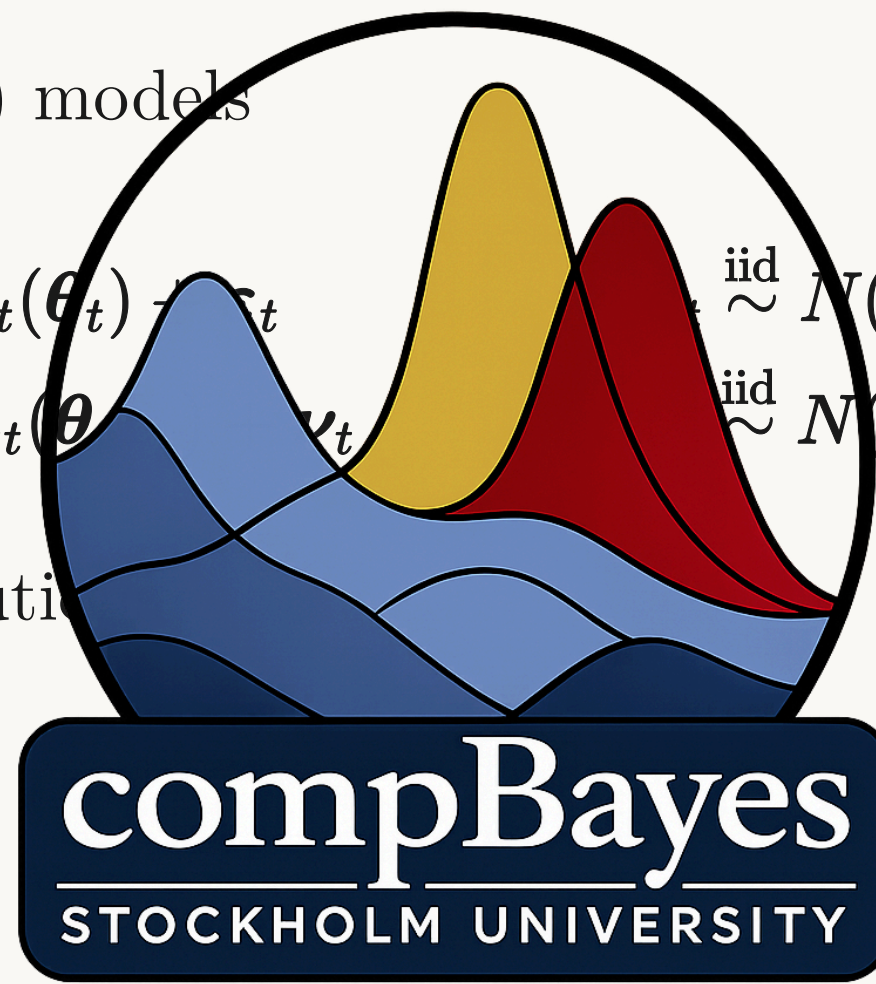
- **Linear Gaussian**

$$\begin{aligned} \mathbf{y}_t &= \mathbf{C}_t \boldsymbol{\theta}_t + \boldsymbol{\varepsilon}_t & \boldsymbol{\varepsilon}_t &\stackrel{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_{\varepsilon,t}) \\ \boldsymbol{\theta}_t &= \mathbf{A}_t \boldsymbol{\theta}_{t-1} + \boldsymbol{\nu}_t & \boldsymbol{\nu}_t &\stackrel{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_{\nu,t}) \end{aligned}$$

- **Non-linear Gaussian** (additive) models

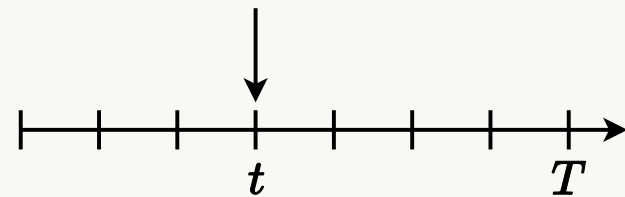
$$\begin{aligned} \mathbf{y}_t &= \mathbf{c}_t(\boldsymbol{\theta}_t) + \boldsymbol{\varepsilon}_t & \boldsymbol{\varepsilon}_t &\stackrel{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_{\varepsilon,t}) \\ \boldsymbol{\theta}_t &= \mathbf{a}_t(\boldsymbol{\theta}_{t-1}) + \boldsymbol{\nu}_t & \boldsymbol{\nu}_t &\stackrel{\text{iid}}{\sim} N(\mathbf{0}, \boldsymbol{\Sigma}_{\nu,t}) \end{aligned}$$

- **Non-Gaussian** - general distribution

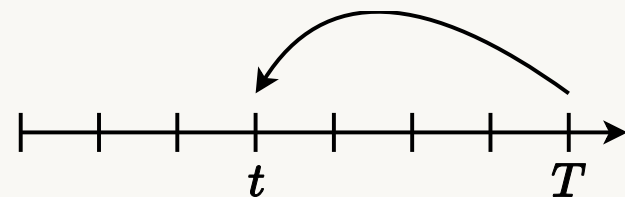


Bayesian inference in state-space models

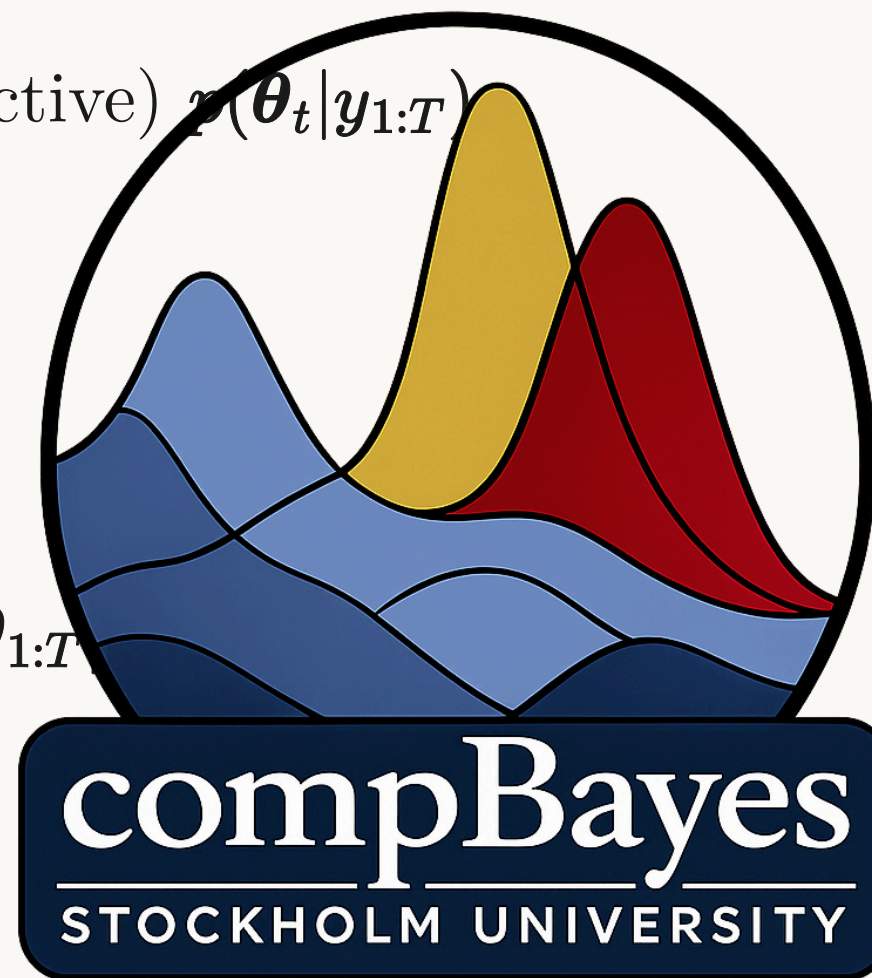
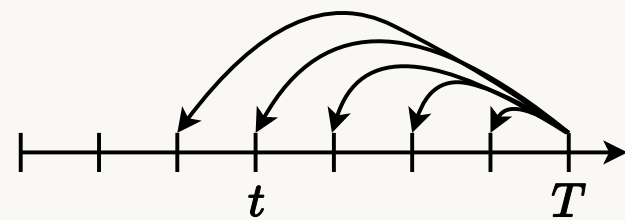
- **Filtering posterior** (instantaneous) $p(\boldsymbol{\theta}_t | y_{1:t})$



- **Smoothing posterior** (retrospective) $p(\boldsymbol{\theta}_t | y_{1:T})$



- **Joint smoothing posterior** $p(\boldsymbol{\theta}_{1:T} | y_{1:T})$



Global-local shrinkage priors - regression [1, 2]

- Linear Gaussian regression with **horseshoe prior**

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma_\varepsilon^2)$$

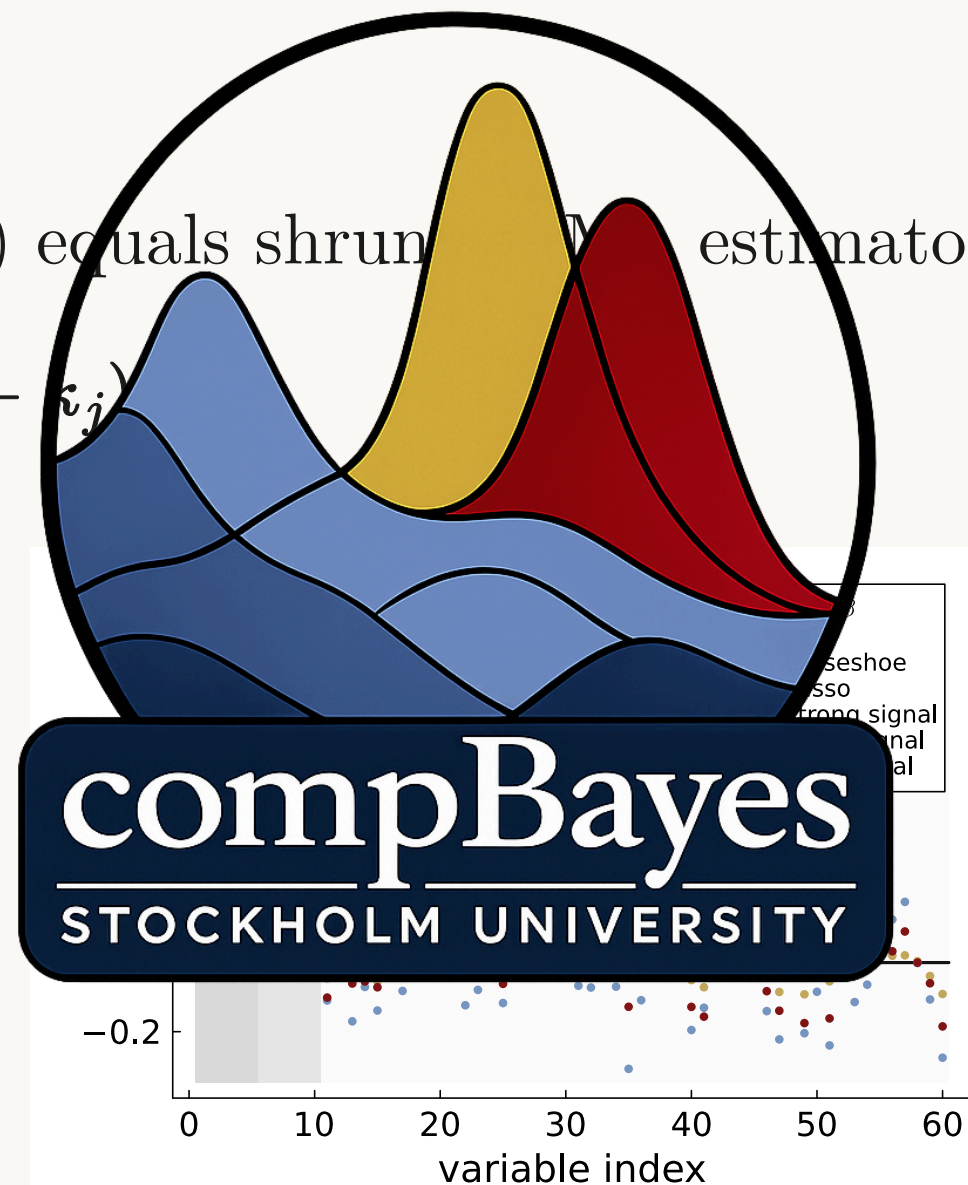
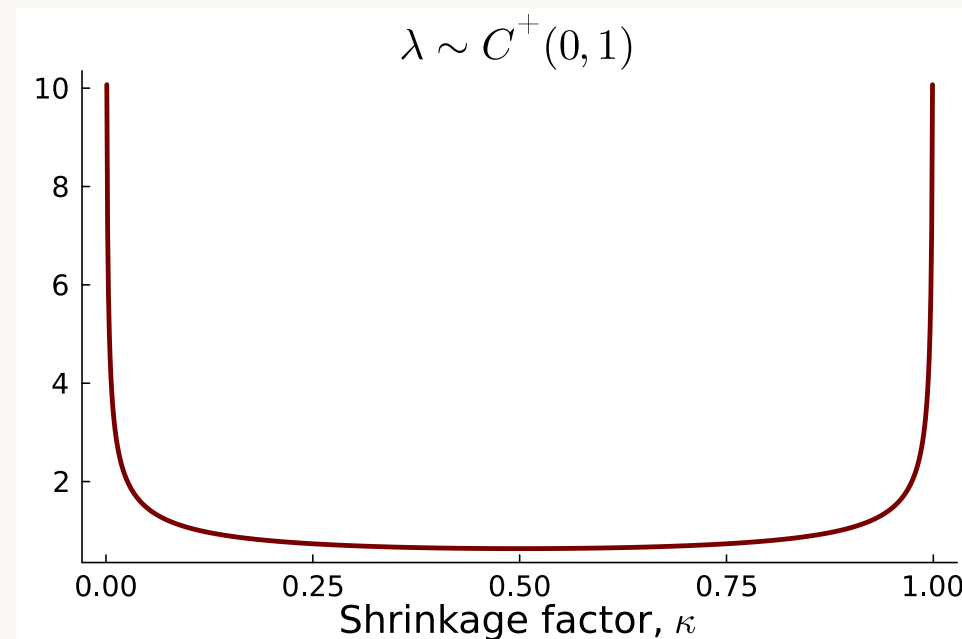
$$\beta_j \stackrel{\text{iid}}{\sim} N(0, \tau^2 \lambda_j^2)$$

$$\tau \sim C^+(0, 1)$$

$$\lambda_j \stackrel{\text{iid}}{\sim} C^+(0, 1)$$

- Posterior mean (for orthogonal $\mathbf{x}$) equals shrunk OLS estimator

$$\tilde{\beta}_j = (1 - \kappa_j) \beta_j$$



Dynamic global-local shrinkage priors for time series

- Local level model - dynamic shrinkage process (DSP) prior

$$y_t = \mu_t + \varepsilon_t$$

$$\mu_t = \mu_{t-1} + \nu_t,$$

$$h_t = \rho h_{t-1} + \eta_t,$$

$$\tau \sim C^+(0, 1)$$

$$\varepsilon_t \stackrel{\text{iid}}{\sim} N(0, \sigma_\varepsilon^2)$$

$$\nu_t \stackrel{\text{ind}}{\sim} N(0, \tau^2 \exp(h_t))$$

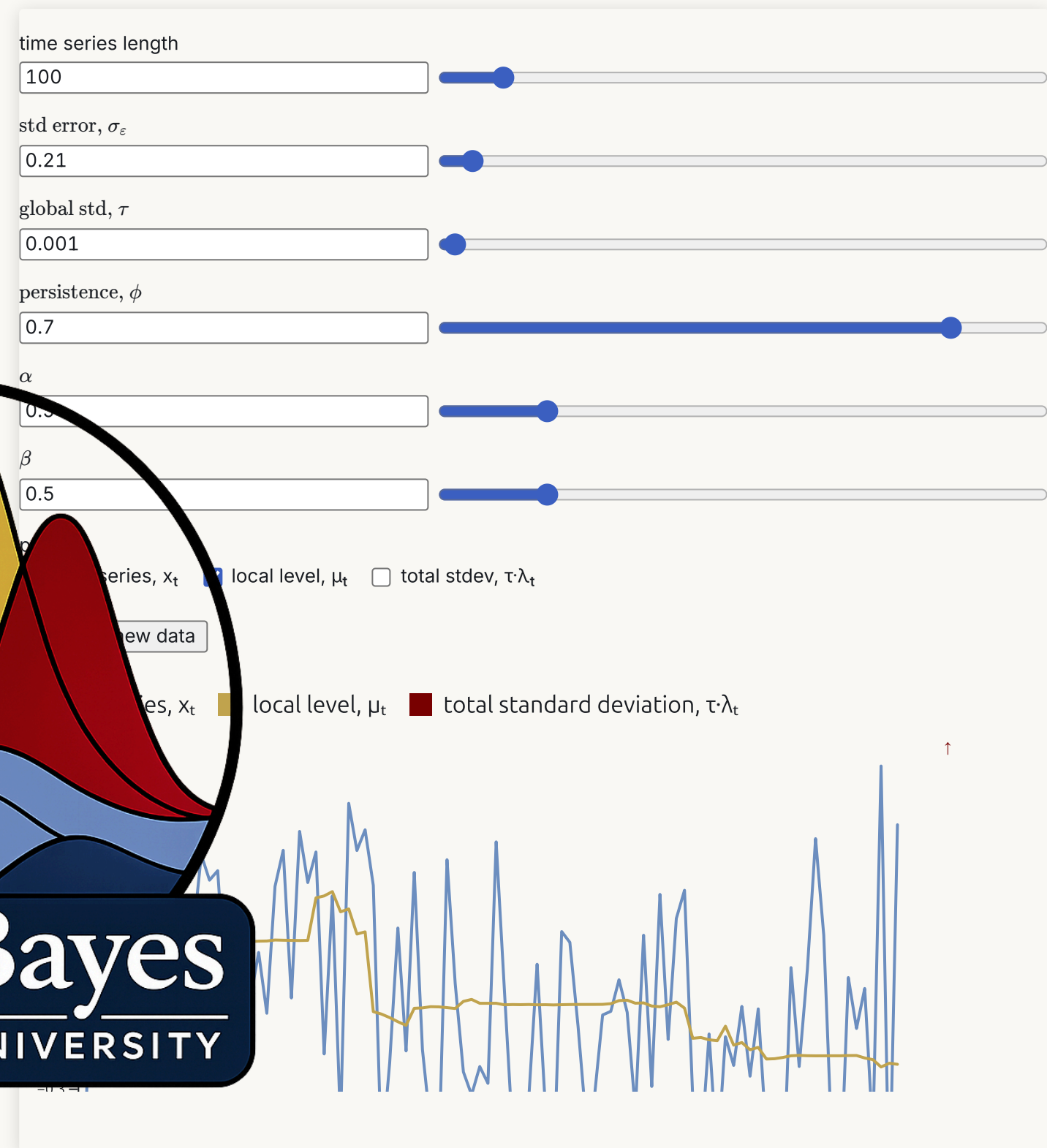
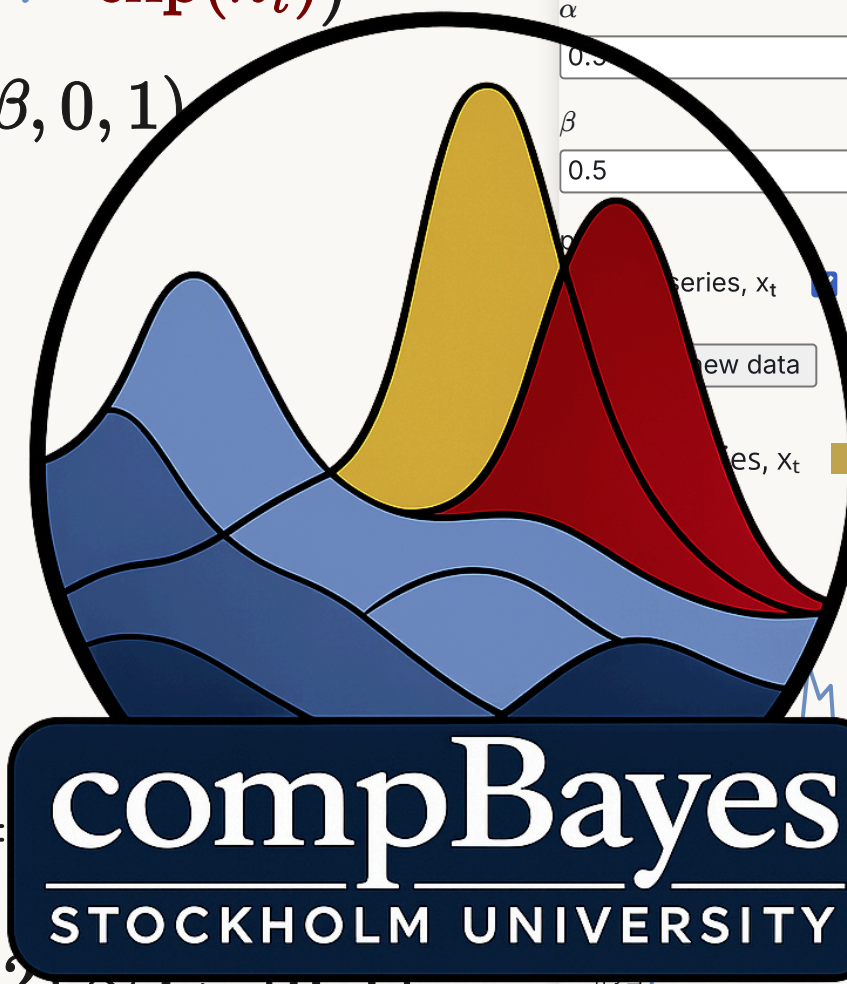
$$\eta_t \stackrel{\text{iid}}{\sim} Z(\alpha, \beta, 0, 1)$$

- Global variance τ^2

- Local variance $\lambda_t^2 = \exp(h_t)$

- DSP becomes horseshoe when $\rho = 0$

$$h_t \sim Z(\alpha, \beta, 0, 1) \iff \lambda_t = \exp(h_t/2) \sim C^+(0, 1)$$



Logistic-Beta distribution (Fisher's Z)

- $X \sim \text{Beta}(\alpha, \beta)$ then

$$Z := \log \left(\frac{X}{1-X} \right) \sim Z(\alpha, \beta, 0, 1)$$

- Location-scale:

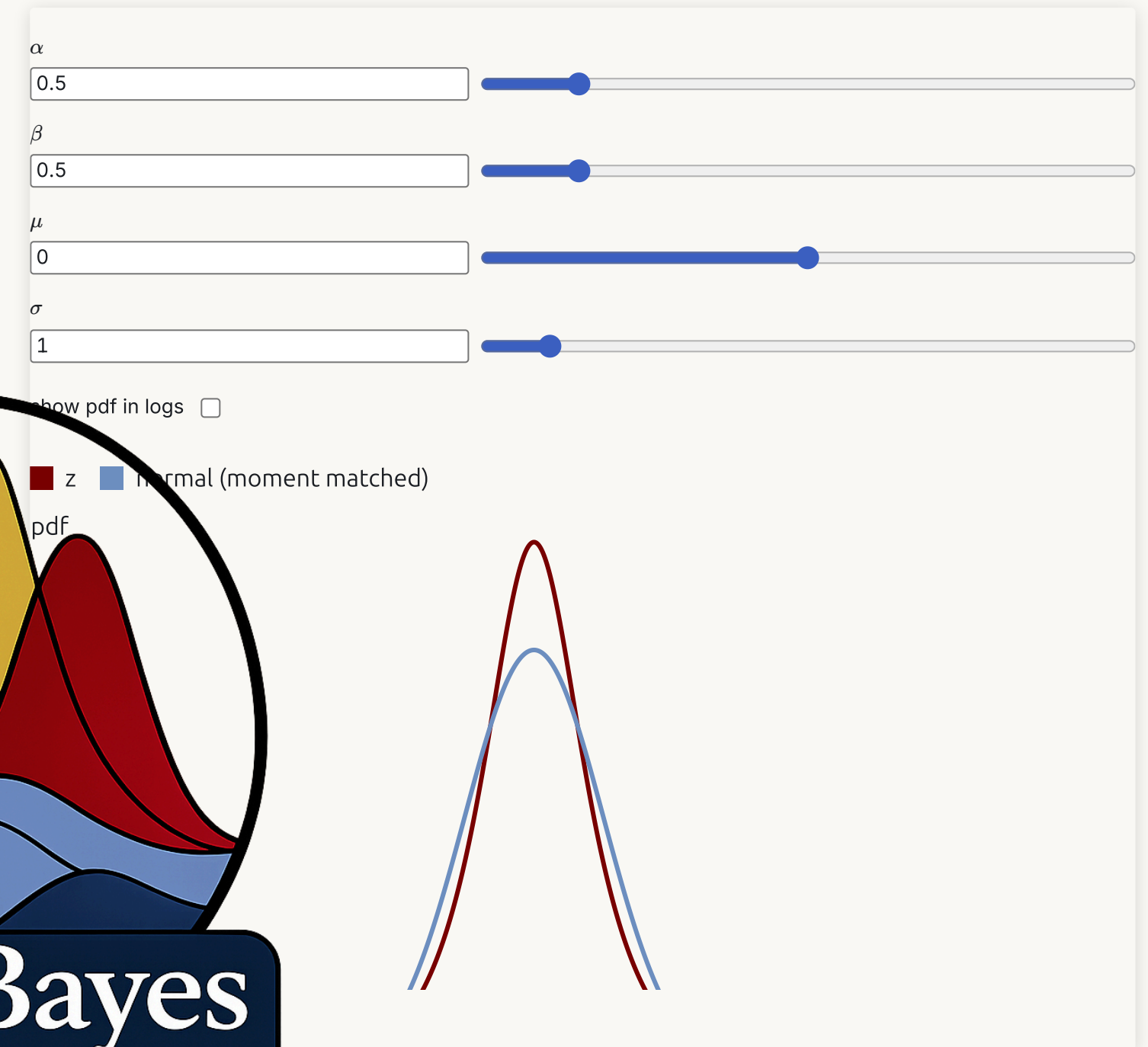
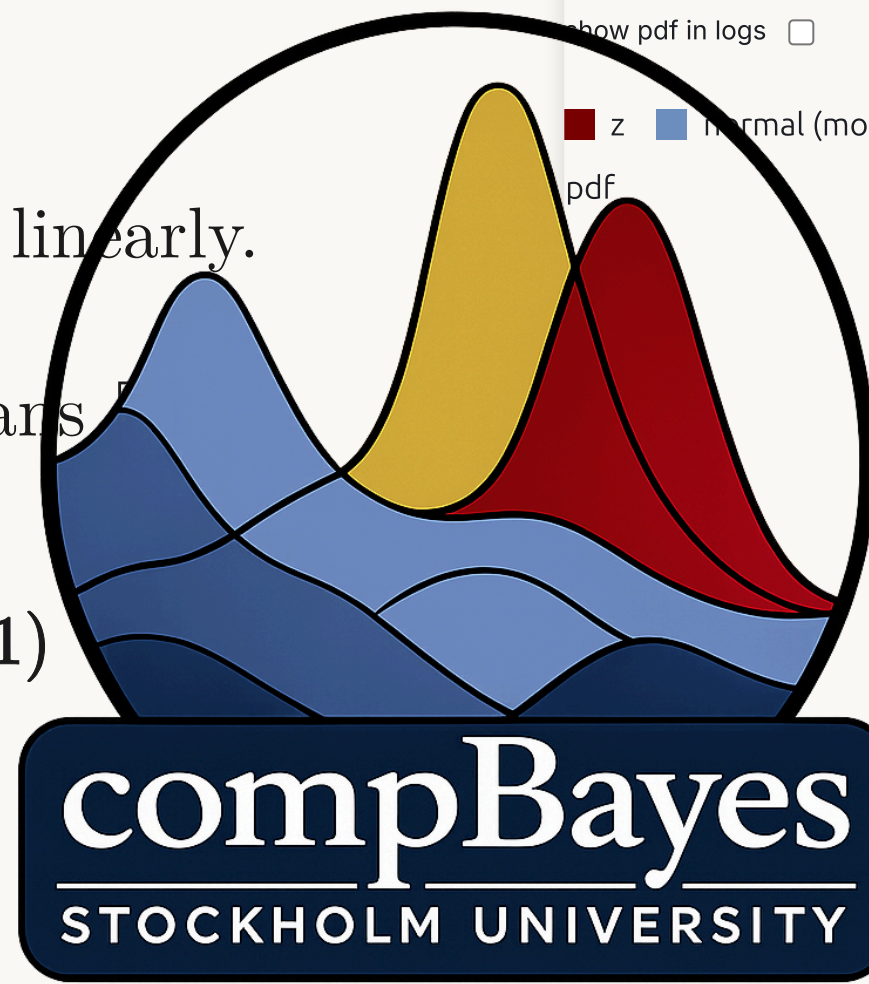
$$\mu + \sigma Z \sim Z(\alpha, \beta, \mu, \sigma)$$

- **Heavy tails** - log density decays linearly.

- Mean-variance mixture of Gaussians

$$Z|\xi \sim N(\xi^{-1}(\alpha - \beta)/2, \xi^{-1})$$

$$\xi \sim \text{PolyaGamma}(\alpha + \beta, 1)$$



Multi-seasonal ARMA with DSP prior [4]

- Seasonal AR with DSP evolution

$$\phi_{p,t}(L)\Phi_{P,t}(L^s)(y_t - \mu_t) = \varepsilon_t, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} N(0, \sigma_t^2)$$

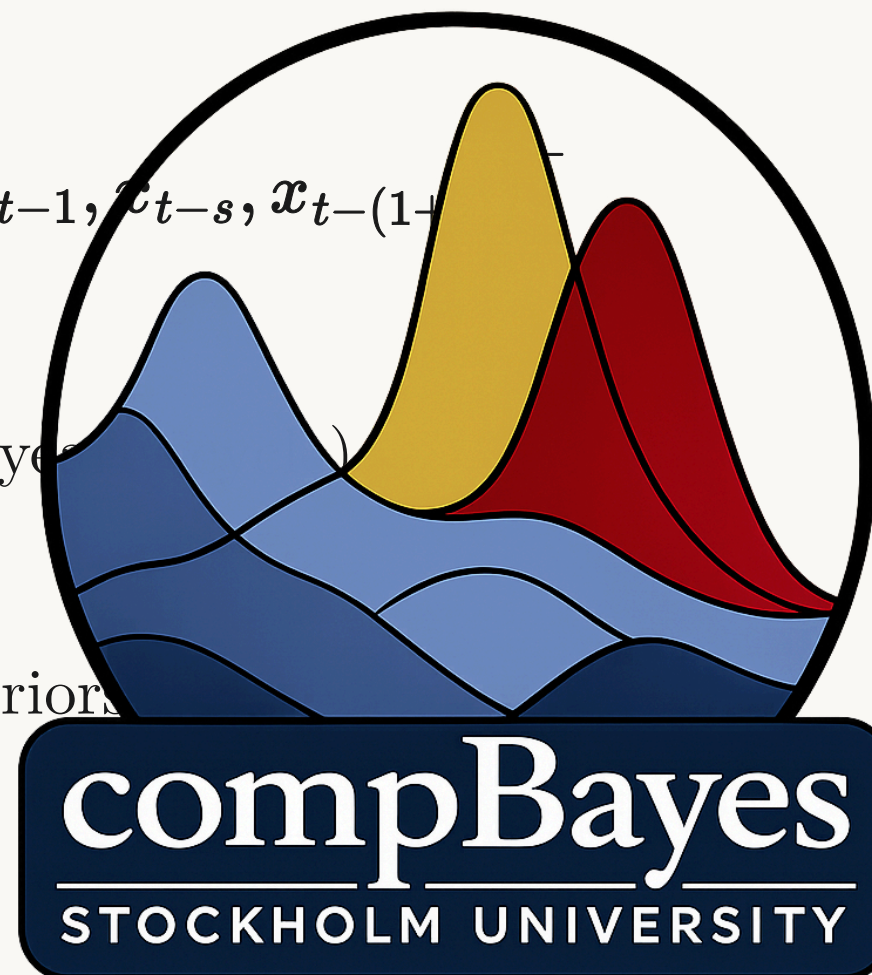
- SAR(1,1) as a **nonlinear Gaussian regression**

$$y_t = \mathbf{z}_t^\top \tilde{\boldsymbol{\phi}} + \varepsilon_t$$

$$\tilde{\boldsymbol{\phi}} = (\phi_1, \Phi_1, -\phi_1 \Phi_1)^\top \text{ and } \mathbf{z}_t = (x_{t-1}, x_{t-s}, x_{t-(1+s)})^\top$$

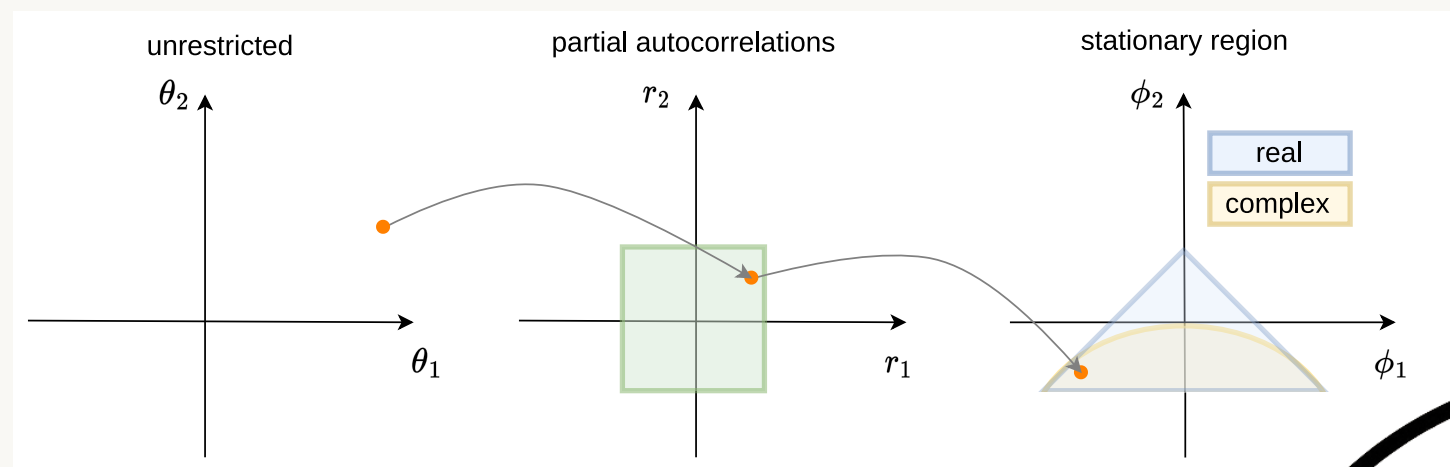
- Extensions

- ✓ **Multi-seasonal** (daily, weekly, yearly)
- ✓ ARMA and Exact likelihood
- ✓ Stochastic volatility with DSP prior
- Multi-seasonal forecasting
- Multi-seasonal VAR



Stability restrictions

Transformation to stability at every t



- Transformation to partial autocorr matters

■ **Monahan**

$$r = \frac{\theta}{\sqrt{1 + \theta^2}}$$

■ **Tanh**

$$r = \frac{e^\theta - e^{-\theta}}{e^\theta + e^{-\theta}}$$

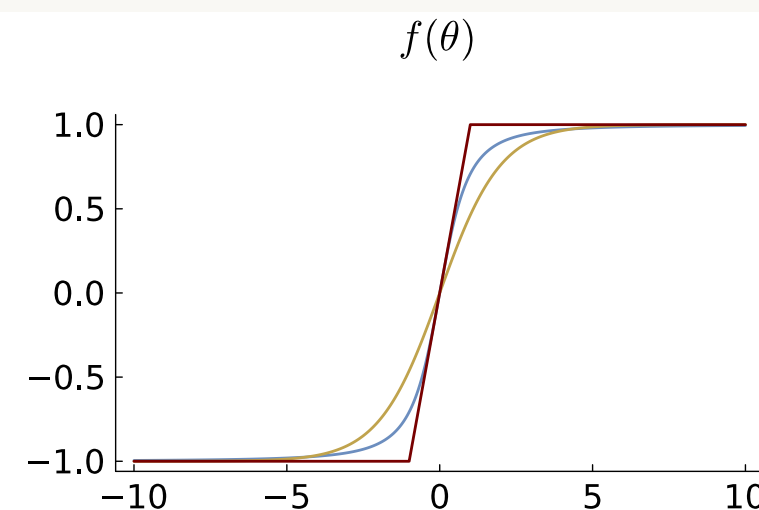
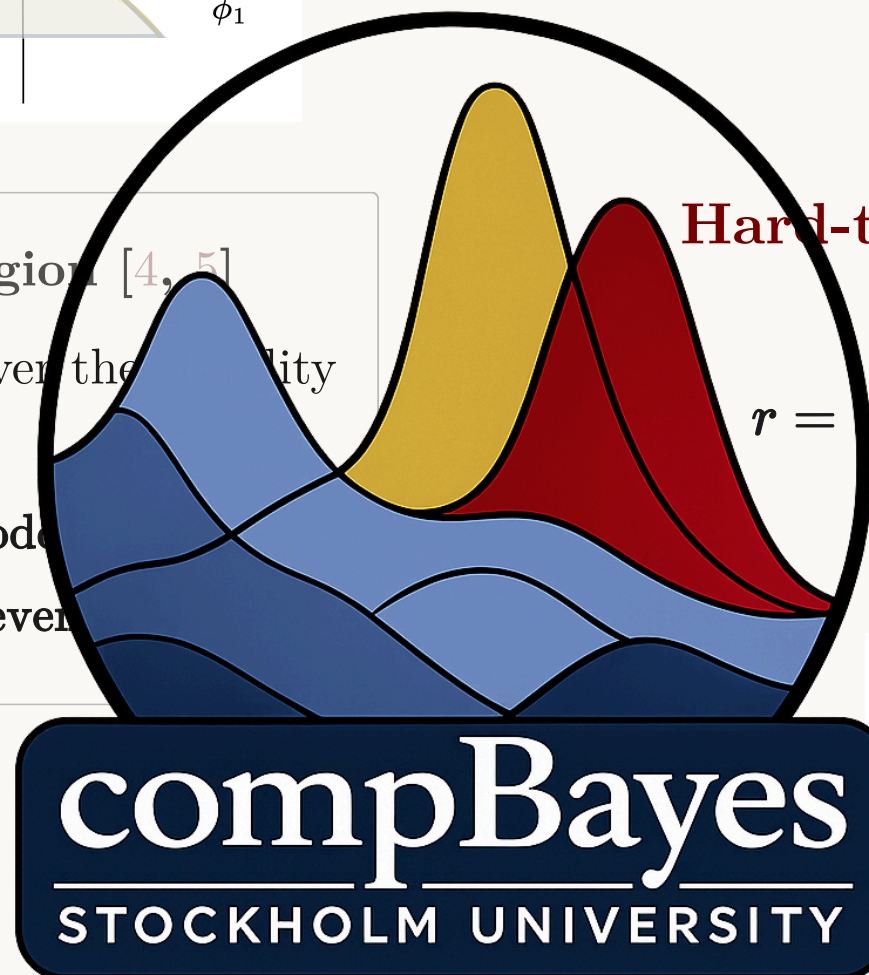
Hard-tanh

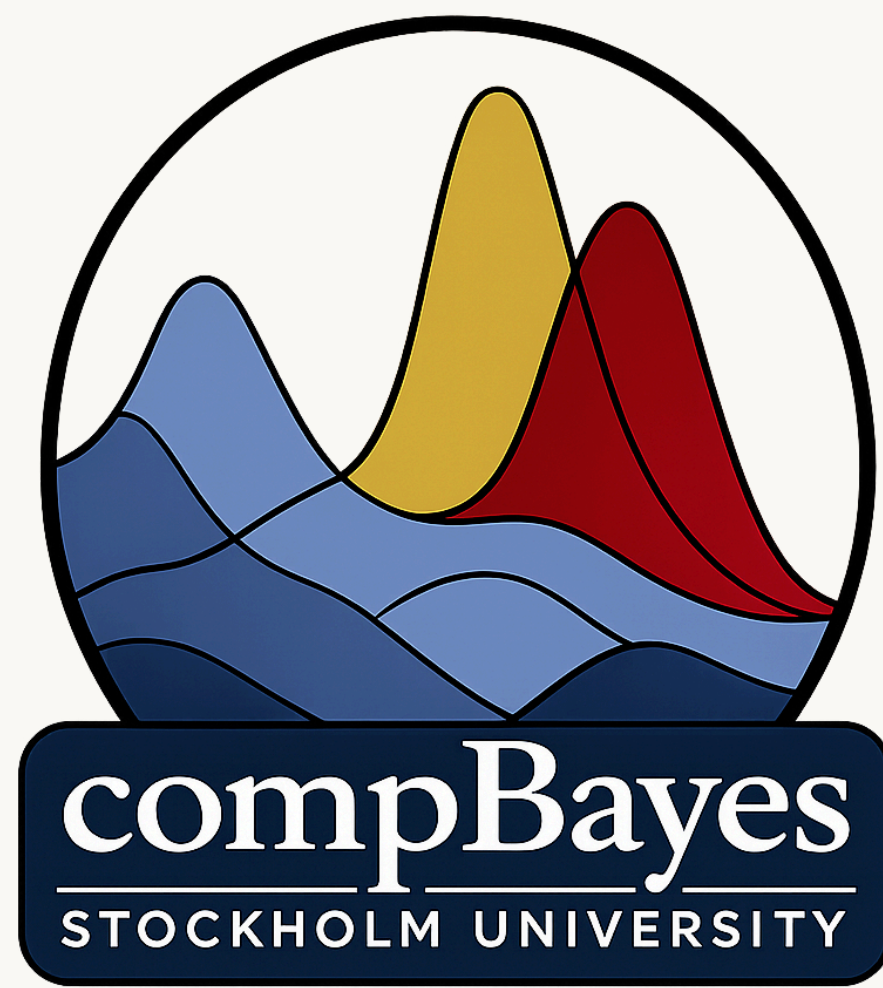
$$r = \begin{cases} -(1 - \epsilon) & \text{if } \theta \leq -(1 - \epsilon) \\ \theta & \text{if } -(1 - \epsilon) < \theta < 1 - \epsilon \\ 1 - \epsilon & \text{if } \theta \geq 1 - \epsilon \end{cases}$$

Lemma - uniform prior over the stability region [4, 5]

The AR parameters ϕ are uniformly distributed over the stability region if, independently,

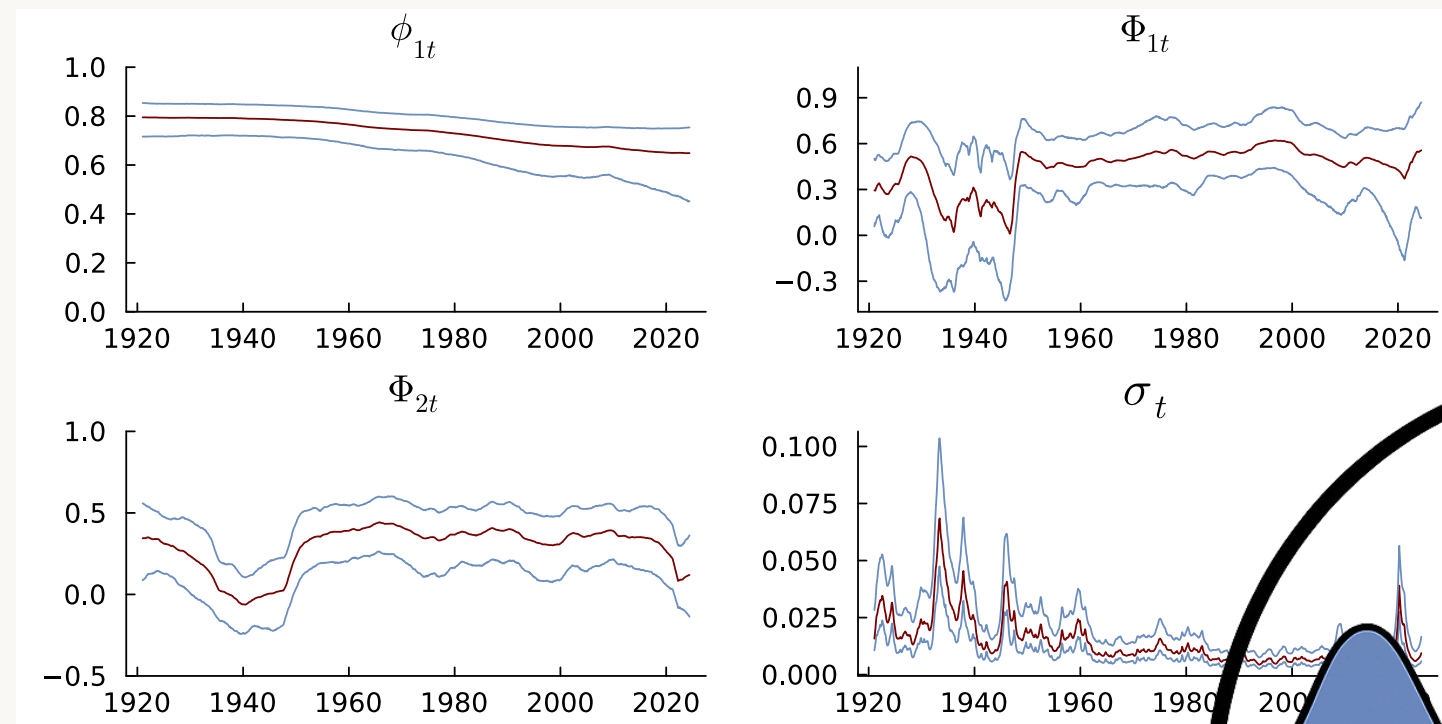
$$\theta_k \sim \begin{cases} t(k+1, 0, \frac{1}{\sqrt{k+1}}) & \text{for } k \text{ is odd} \\ t_{\text{skew}}(\frac{k}{2}, \frac{k+2}{2}, 0, \frac{1}{\sqrt{k+1}}) & \text{for } k \text{ is even} \end{cases}$$



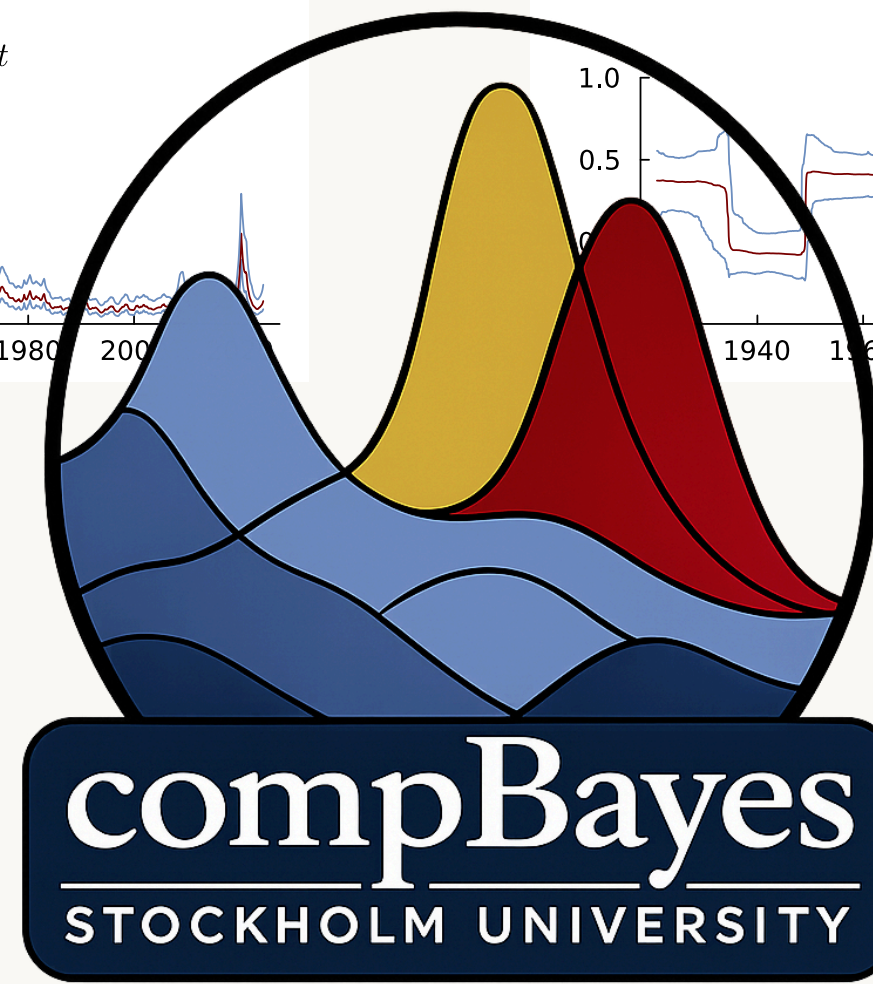
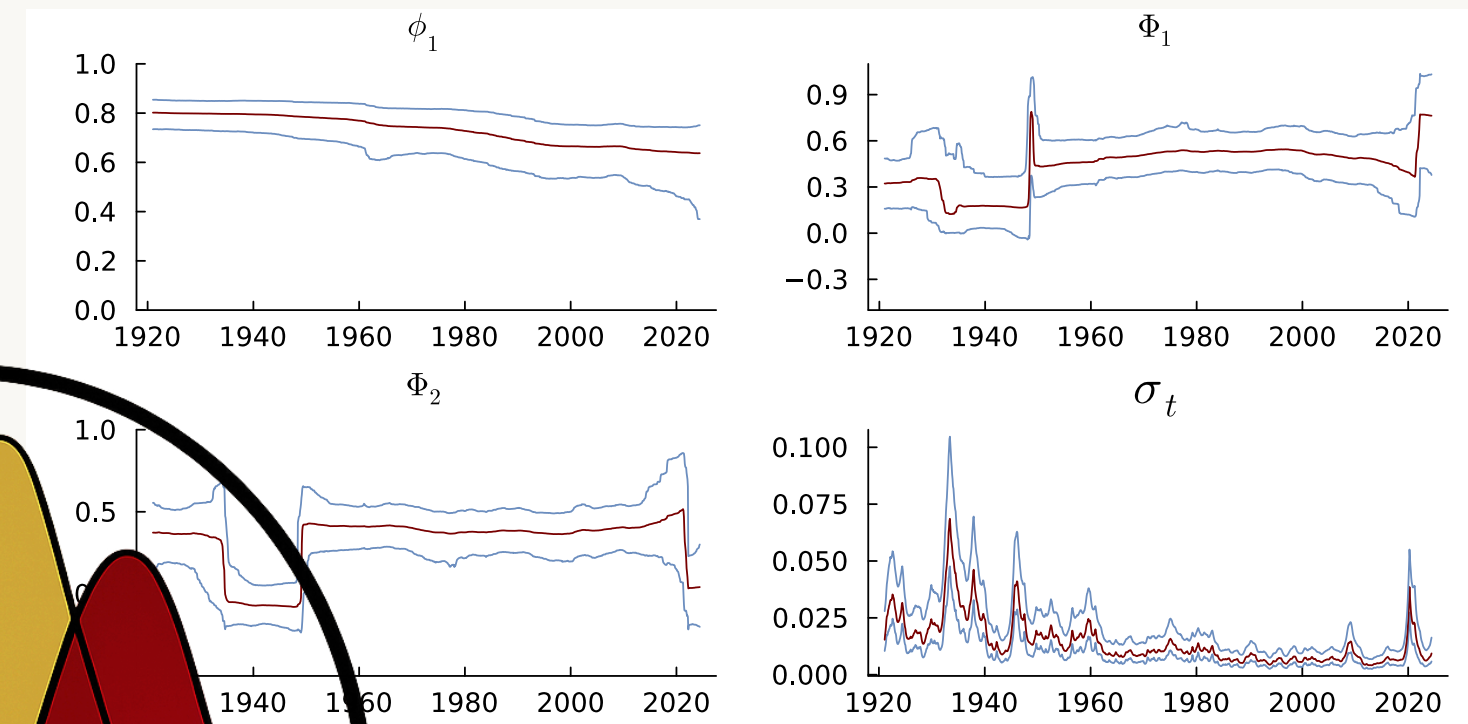


Seasonal AR with DSP prior - US industrial production 1919-2024

SAR(1,2) - Homoscedastic Gaussian



SAR(1,2) - DSP



Time-varying generalized linear models with DSP evolution

- **GLMs** with time-varying parameters

<detached>

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$$y_t | \mathbf{x}_t \sim \text{ExpFamily} \left(k^{-1}(\mathbf{x}_t^\top \boldsymbol{\beta}_t), \varphi_t \right)$$

$$\boldsymbol{\beta}_t = \boldsymbol{\beta}_{t-1} + \boldsymbol{\nu}_t, \quad \boldsymbol{\nu}_t \stackrel{\text{iid}}{\sim} N(\mathbf{0}, \text{Diag}(\exp(\boldsymbol{\tau}^2 \odot \mathbf{h}_t)))$$

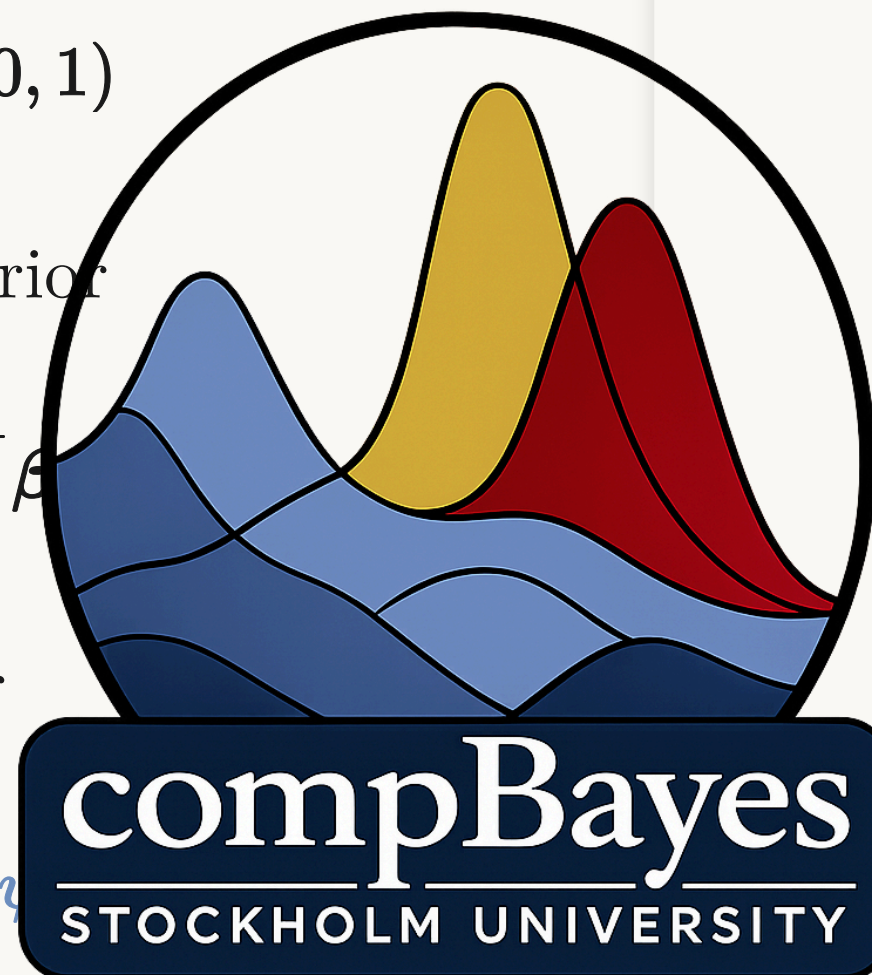
$$\mathbf{h}_t = \Phi \mathbf{h}_{t-1} + \boldsymbol{\eta}_t, \quad \eta_{tk} \stackrel{\text{iid}}{\sim} Z(\alpha, \beta, 0, 1)$$

- **Poisson regression** with DSP prior

$$y_t | \mathbf{x}_t \sim \text{Pois}(\mu_t = \exp(\mathbf{x}_t^\top \boldsymbol{\beta}_t))$$

- **Beta regression** with DSP prior

$$y_t | \mathbf{x}_t, z_t \sim \text{Beta}(\mu_t = \text{logistic}(\mathbf{x}_t^\top \boldsymbol{\beta}_t), \nu_t)$$

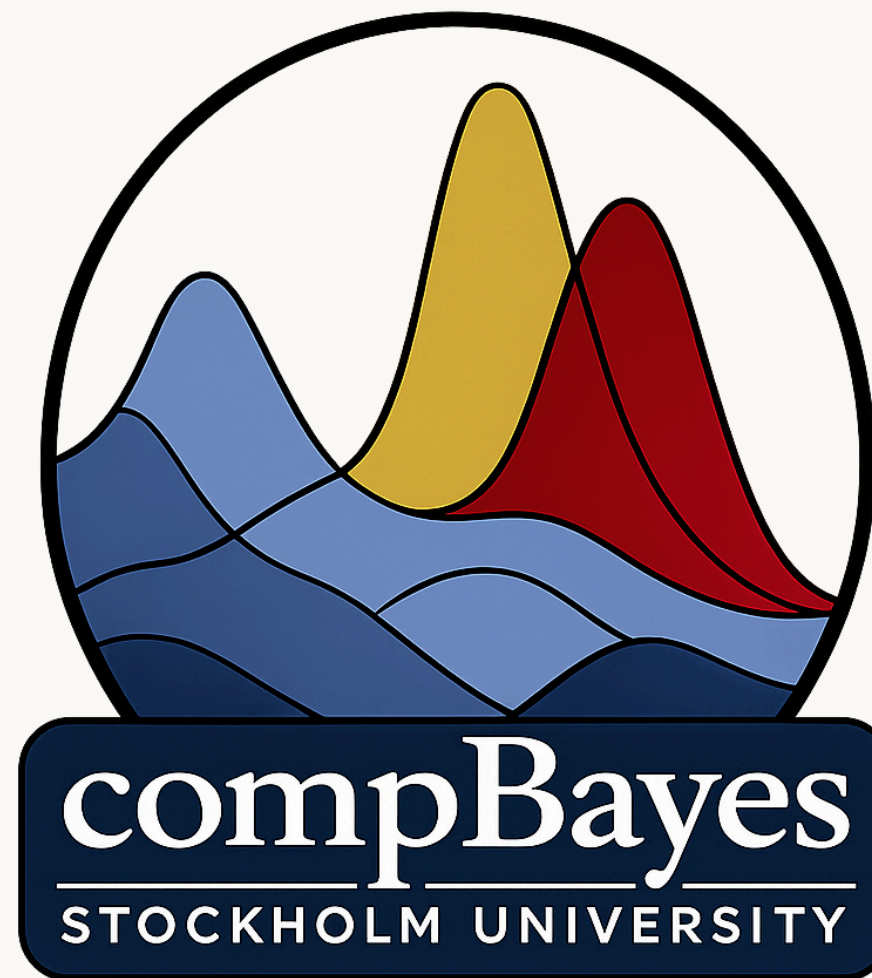


Bayesian inference in state-space models with DSP prior

- Joint smoothing posterior ($\boldsymbol{\psi}$ holds all static parameters):

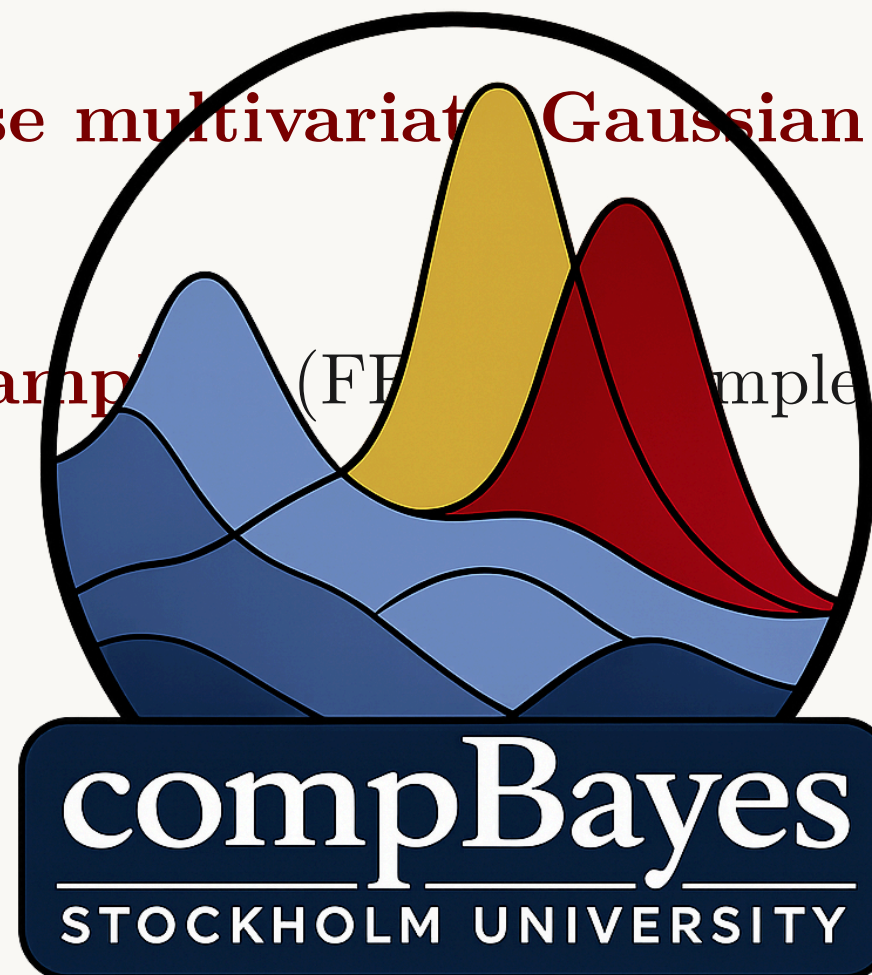
$$p(\boldsymbol{\theta}_{1:T}, \boldsymbol{h}_{1:T}, \boldsymbol{\psi} | \boldsymbol{y}_{1:T})$$

- Gibbs sampling/MCMC
- Particle MCMC
- Hamiltonian Monte Carlo (HMC)
- Variational approximations
- Integrated Nested Laplace Approximation (INLA)?



Gibbs sampling in linear Gaussian model with DSP prior

- **Polya-Gamma augmentation** to turn Z -distribution into conditionally Gaussian. [6, 7]
- **Square-and-log trick** to turn variance $\exp(h_t)$ into a mean h_t . [8]
- Fast sampling of $\mathbf{h}_{1:T}$ using **sparse multivariate Gaussian** [9].
- **Forward-filtering-backward sampling** (FFBS) to sample $\boldsymbol{\theta}_{1:T}$ [10, 11]



Gaussian approximation filters

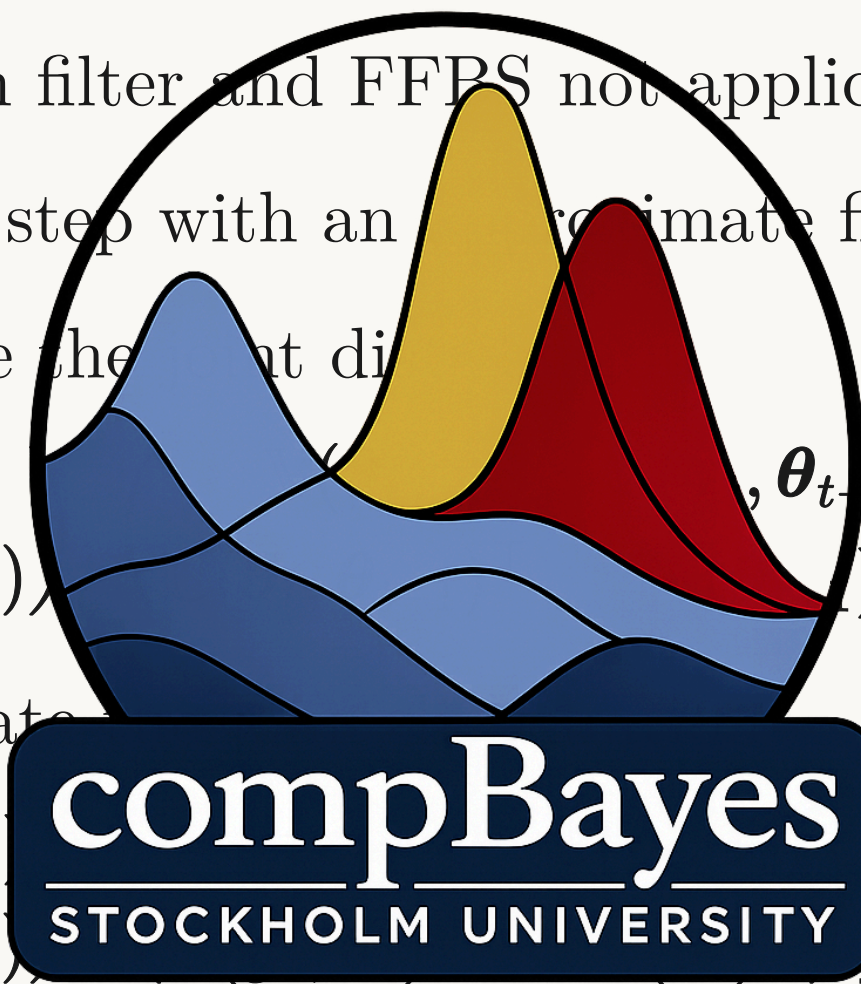
- **Forward-filtering-backward sampling** for linear Gaussian models [10, 11]
 - Run Kalman filter $p(\boldsymbol{\theta}_t | \mathbf{y}_{1:t})$ forward for $t = 1, 2, \dots, T$
 - Sample $p(\boldsymbol{\theta}_t | \boldsymbol{\theta}_{t+1}, \mathbf{y}_{1:T})$ backward for $t = T, T-1, \dots, 1$
- Non-linear/non-Gaussian: Kalman filter and FFBS not applicable.
- General idea: replace the filtering step with an approximate filter.

- **Prior propagation:** approximate the joint distribution

$$\begin{pmatrix} \boldsymbol{\theta}_t \\ \boldsymbol{\theta}_{t-1} \end{pmatrix} | \mathbf{y}_{1:t-1} \stackrel{\text{approx}}{\sim} N \left[\begin{pmatrix} \mathbb{E}(\boldsymbol{\theta}_t) \\ \mathbb{E}(\boldsymbol{\theta}_{t-1}) \end{pmatrix}, \begin{pmatrix} \text{Cov}(\boldsymbol{\theta}_t, \boldsymbol{\theta}_{t-1}) \\ \text{Cov}(\boldsymbol{\theta}_{t-1}, \boldsymbol{\theta}_t) \end{pmatrix} \right]$$

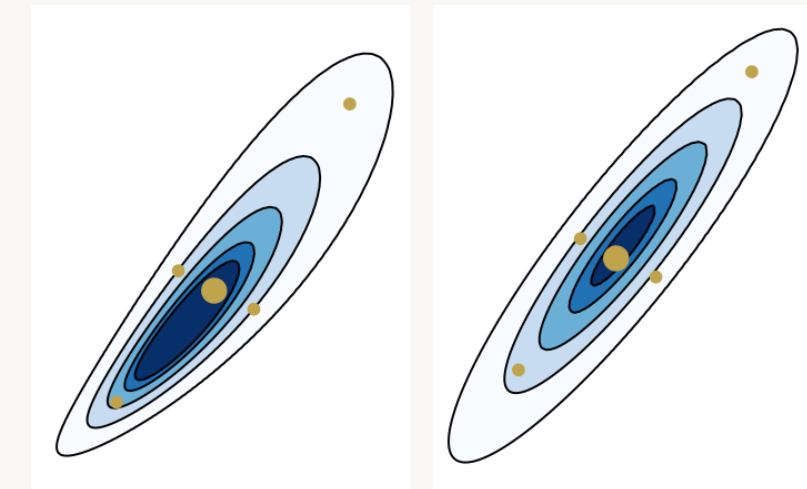
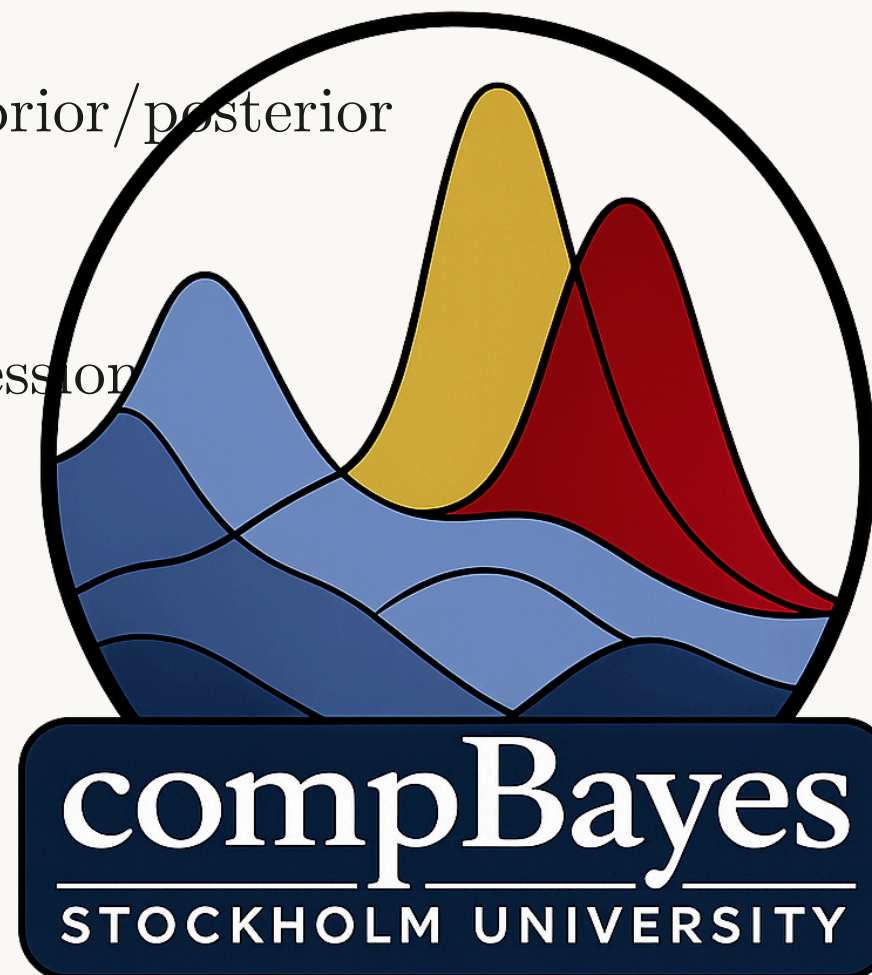
- **Observation update:** approximate

$$\begin{pmatrix} \mathbf{y}_t \\ \boldsymbol{\theta}_t \end{pmatrix} | \mathbf{y}_{1:t-1} \stackrel{\text{approx}}{\sim} N \left[\begin{pmatrix} \mathbb{E}(\mathbf{y}_t) \\ \mathbb{E}(\boldsymbol{\theta}_t) \end{pmatrix}, \begin{pmatrix} \text{Cov}(\mathbf{y}_t, \boldsymbol{\theta}_t) \\ \text{Cov}(\boldsymbol{\theta}_t, \mathbf{y}_t) \end{pmatrix} \right]$$

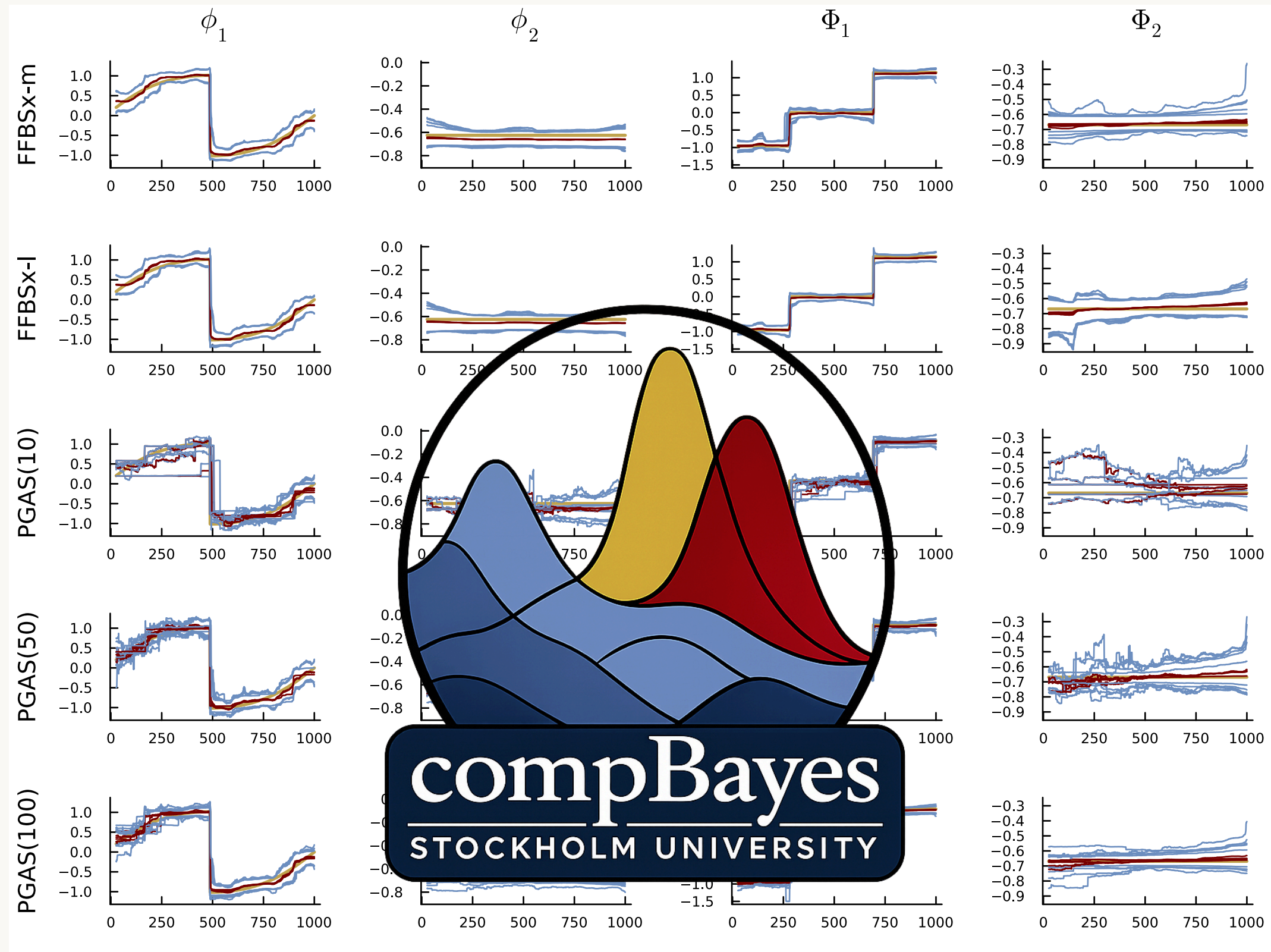


Gaussian approximation filters

- **Extended Kalman filter**
 - Linearize measurement/transition models
 - Apply standard Kalman filter
- **Unscented Kalman filter**
 - Propagate sigma point approx of prior/posterior
- **Prior linearization filter**
 - Linearize by statistical linear regression



Simulated example SAR(2,2) - MCMC convergence



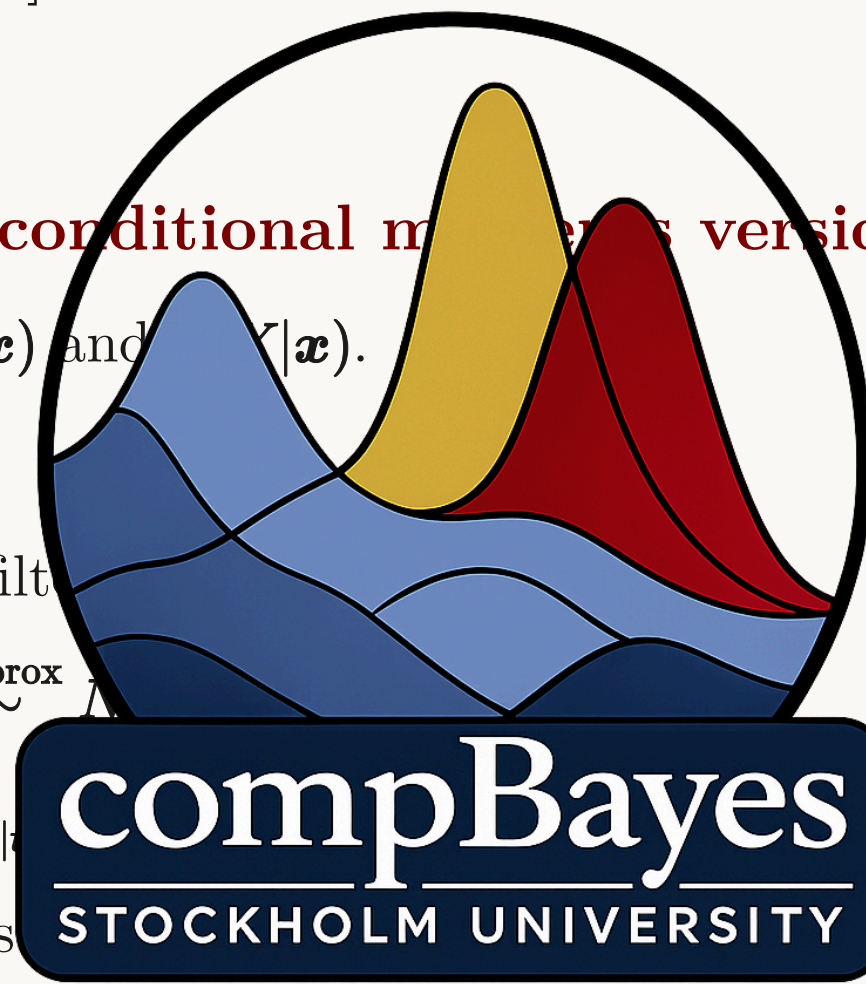
Iterated Gaussian approximation filters

- Nonlinear Gaussian models
 - **Iterated extended Kalman filter** [12]
 - **Iterated unscented Kalman filter** [13]
 - **Posterior linearization filter** [14]
- Non-Gaussian models
 - **Posterior linearization filter - conditional modes version** [15]
 - Statistical linearization using $\mathbb{E}(Y|\mathbf{x})$ and $\mathbb{V}(Y|\mathbf{x})$.
 - **Laplace filter** [16, 17]
 - Direct Gaussian approximation of filter

$$\boldsymbol{\theta}_t | y_{1:t} \approx$$

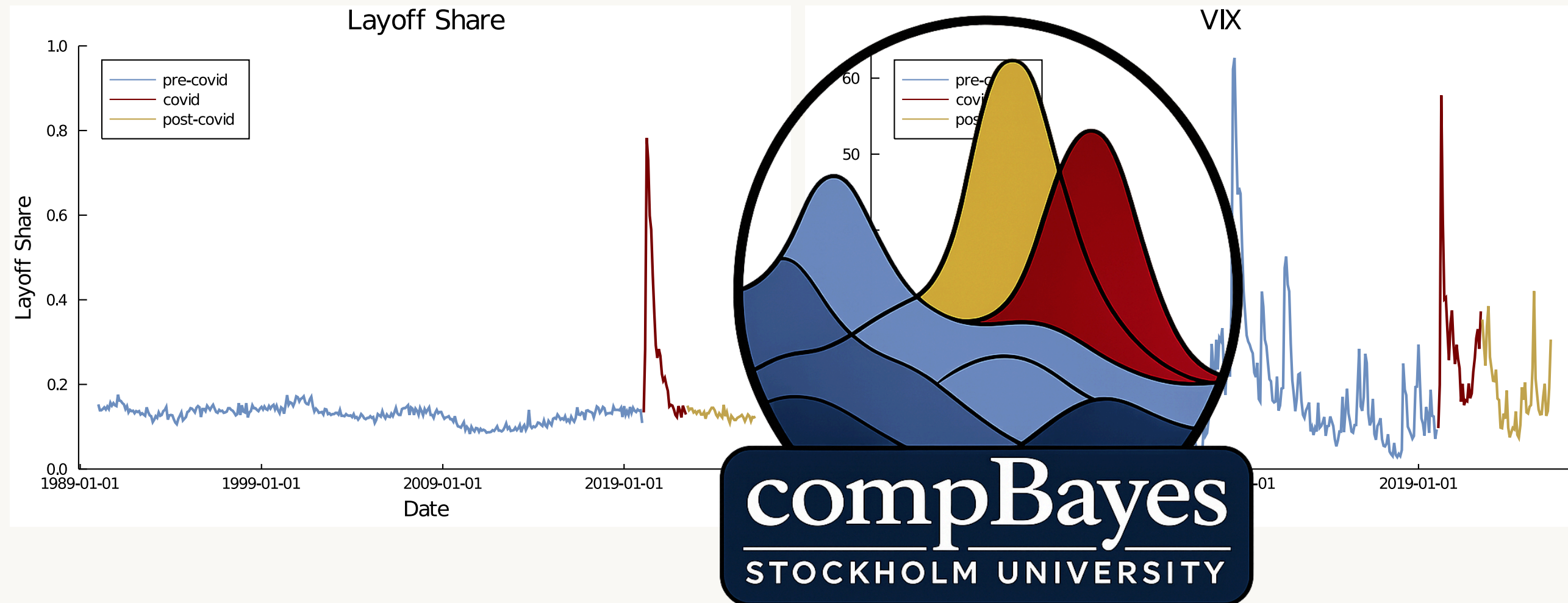
where $\tilde{\boldsymbol{\theta}}_{t|t}$ is posterior mode and $\tilde{\boldsymbol{\Sigma}}_{t|t}$ is posterior covariance.

- **Automatic differentiation** makes

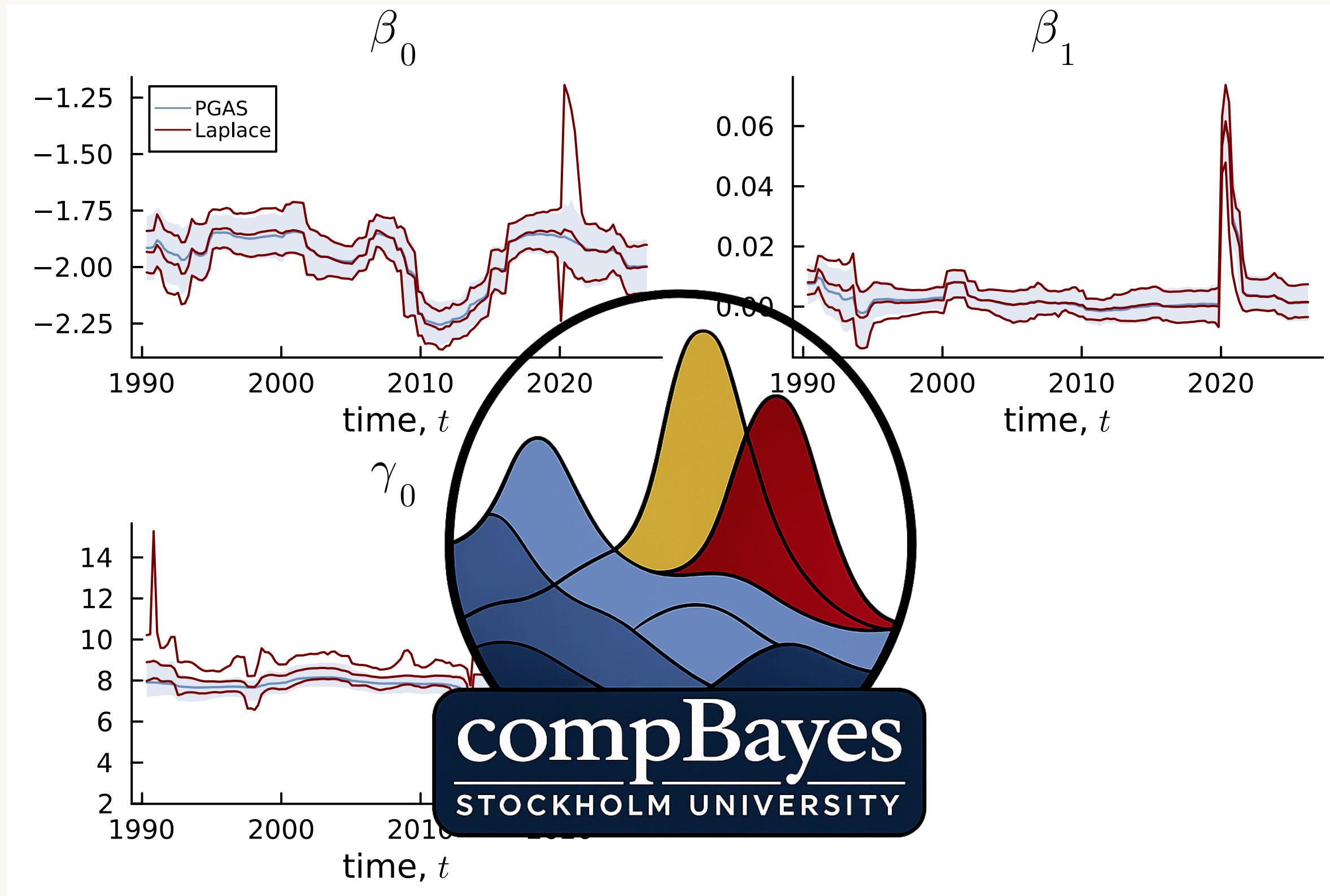


Time-varying Beta regression for US layoff proportion

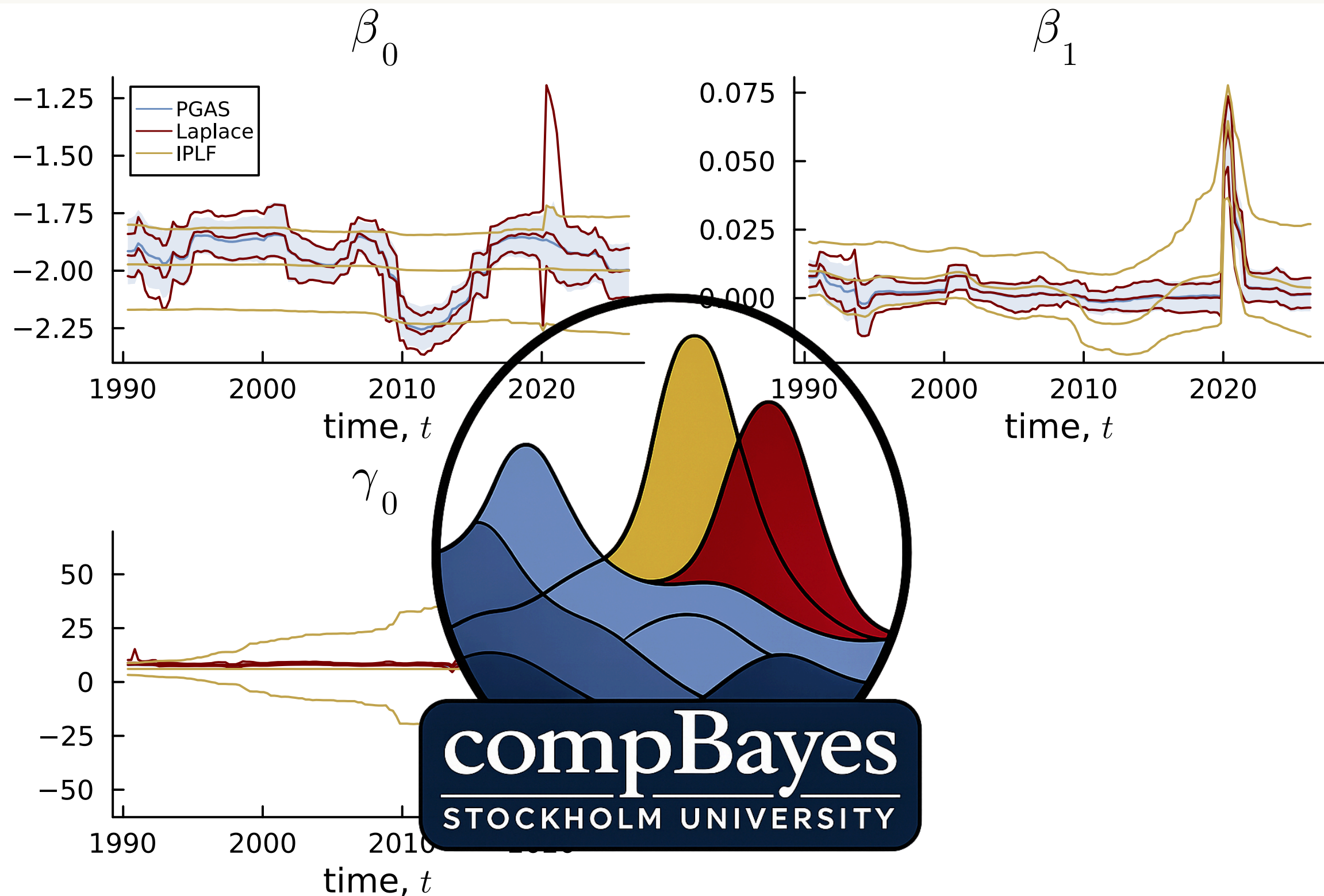
- Monthly US data from 1990:1-2026:4 on:
 - **layoff** - job losers on layoff as % of total unemployed
 - **vix** - CBOE Volatility Index (“fear index”)



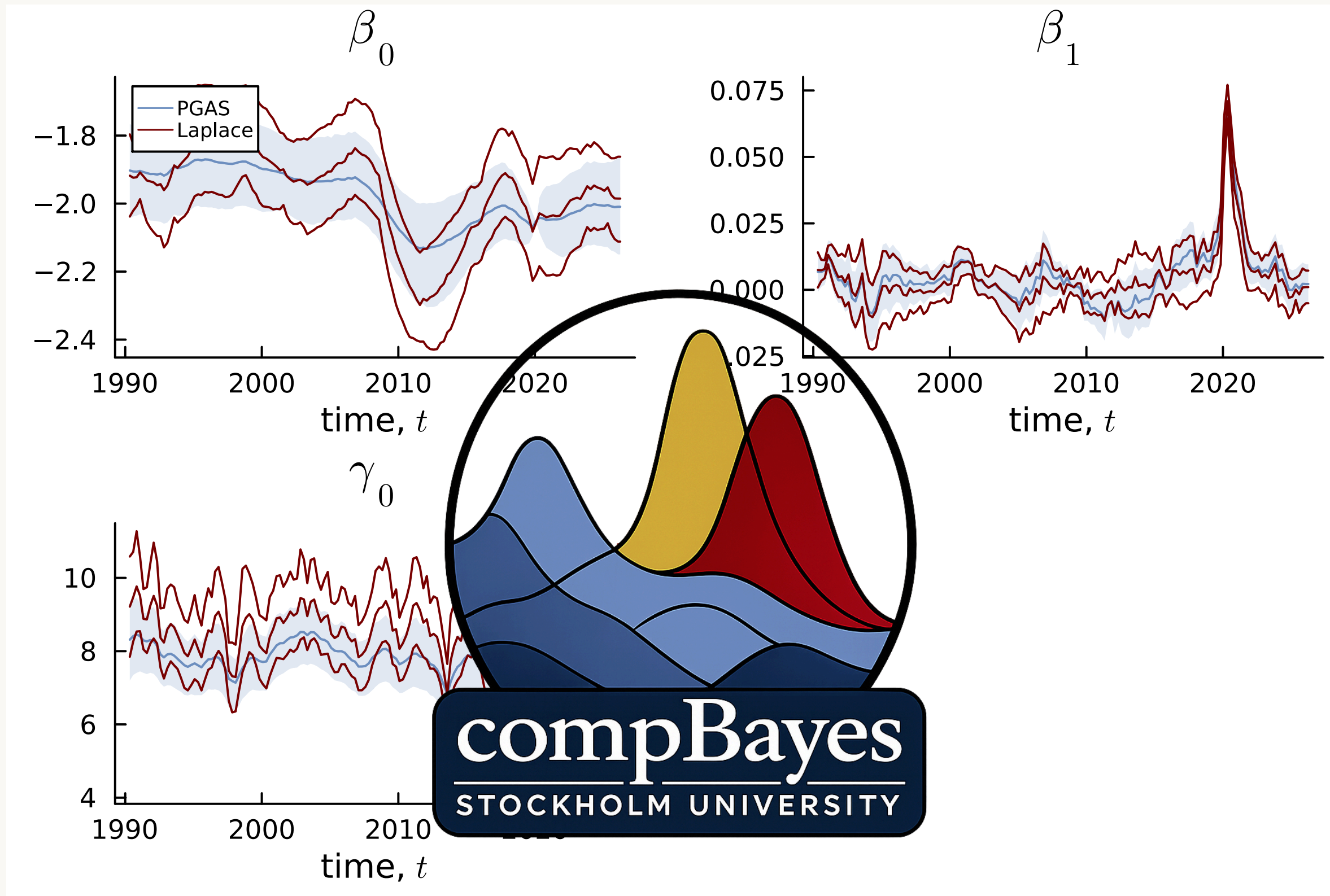
Beta regression for US layoff proportion - dsp prior



Linear approximations struggle with Beta regression

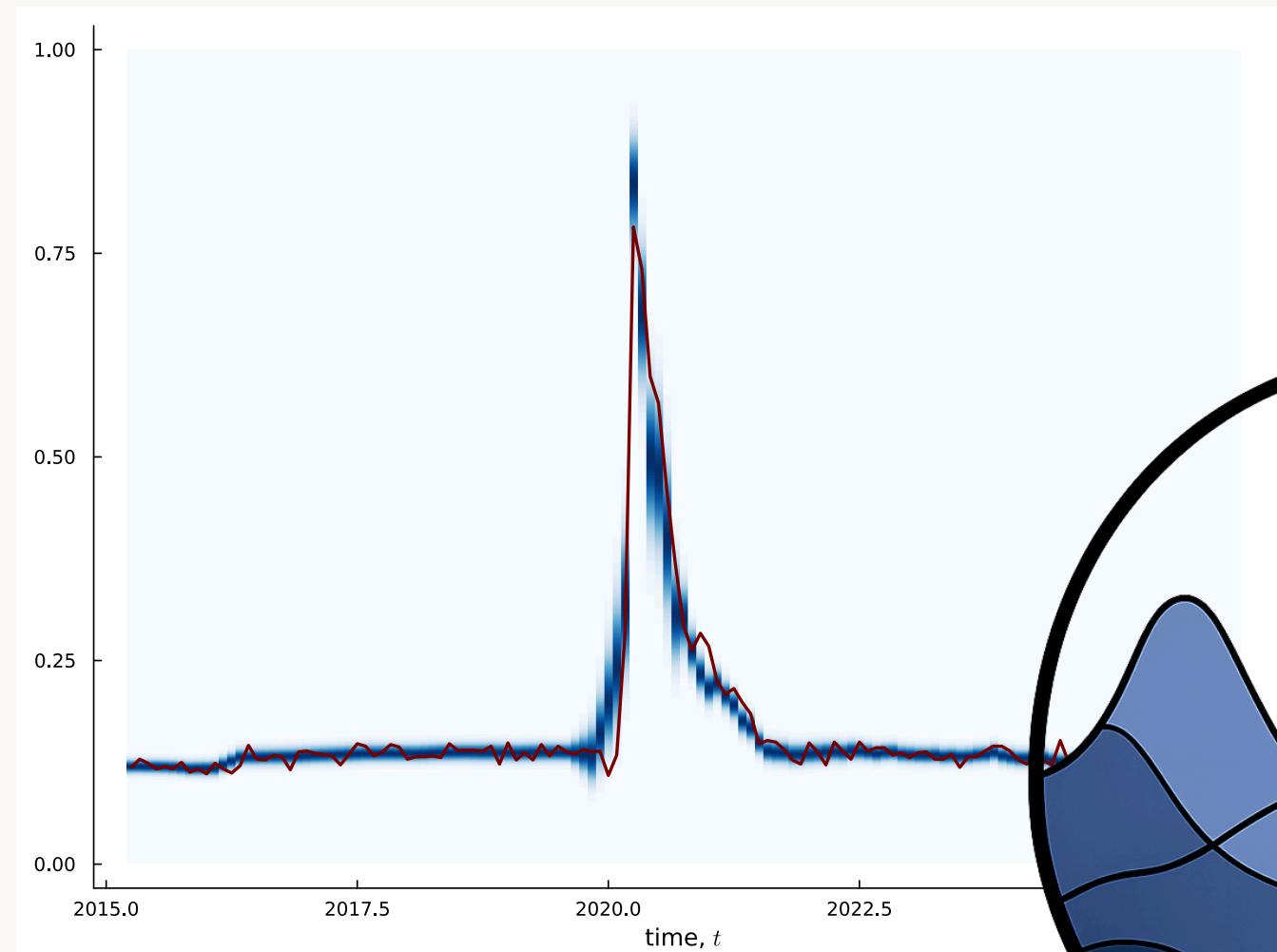


Beta regression for US layoff proportion - homoscedastic Gauss prior

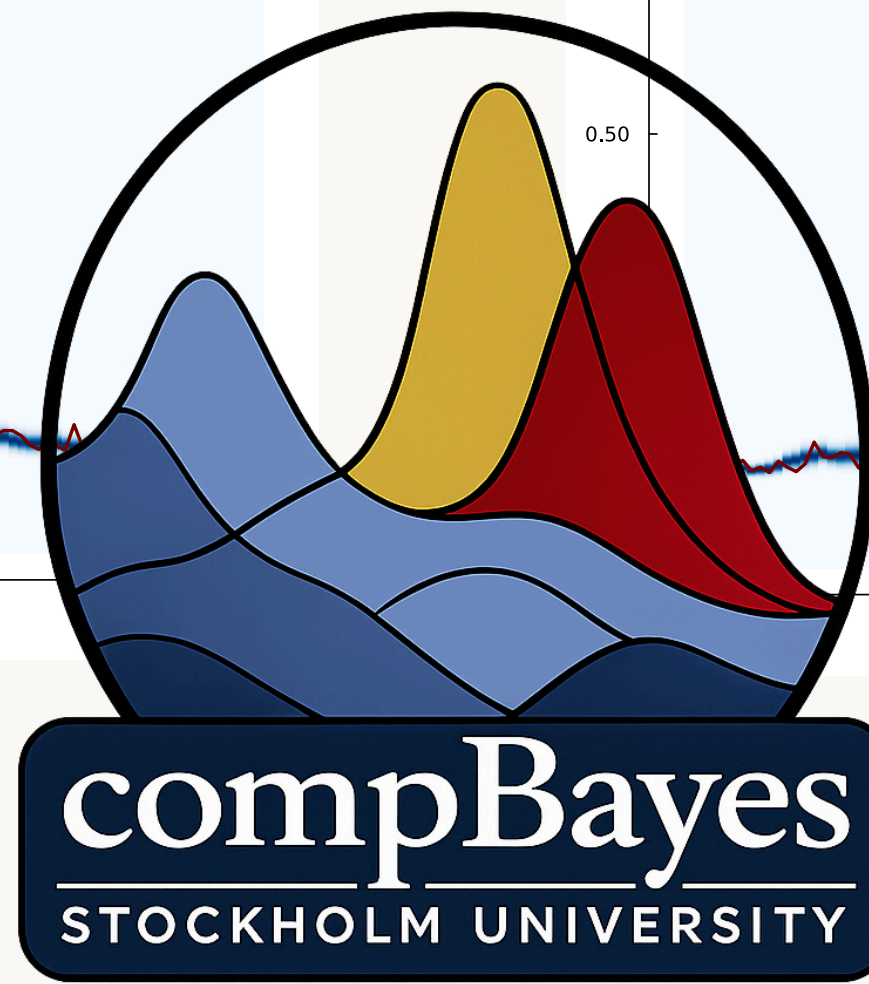
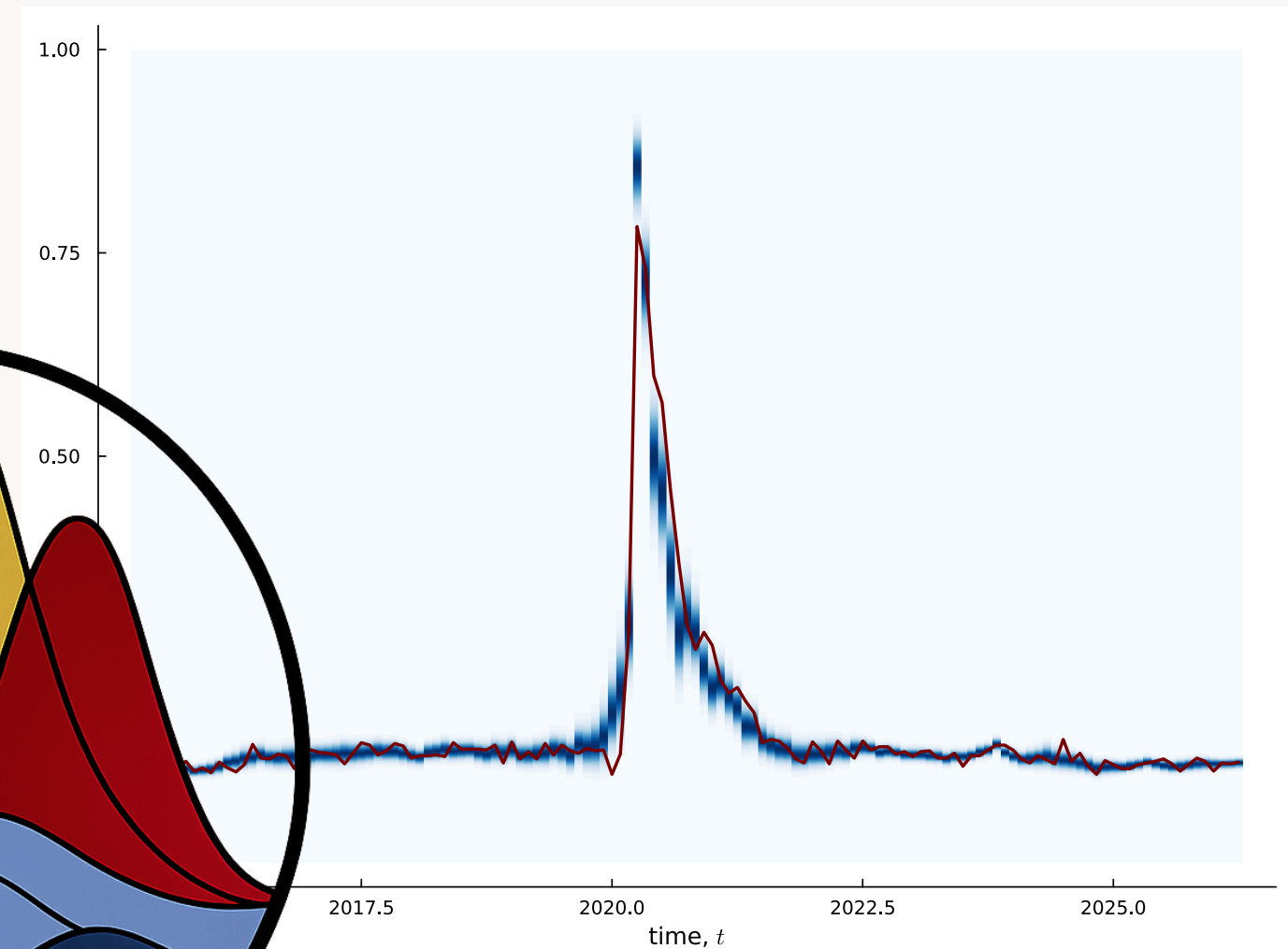


Beta regression for US layoff - fit at posterior median around pandemic

DSP



Homoscedastic Gaussian



Grouping observations for speed and Gaussianity

- Gaussian approximations at every time t .
- How Gaussian are the **likelihood contributions** $p(y_t|\boldsymbol{\theta}_t)$ in $\boldsymbol{\theta}_t$?
- No asymptotic Bernstein-von Mises effect to help.
- A tight propagated prior $\boldsymbol{\theta}_t \sim N(\hat{\boldsymbol{\theta}}_{t|t-1}, \Omega_{t|t-1})$ helps a lot.
- But DSP can give large innovations, so $\Omega_{t|t-1}$ can be large.
- **Group observations** in G groups of observations in group g .
- **Same state $\boldsymbol{\theta}_g$ for observations**

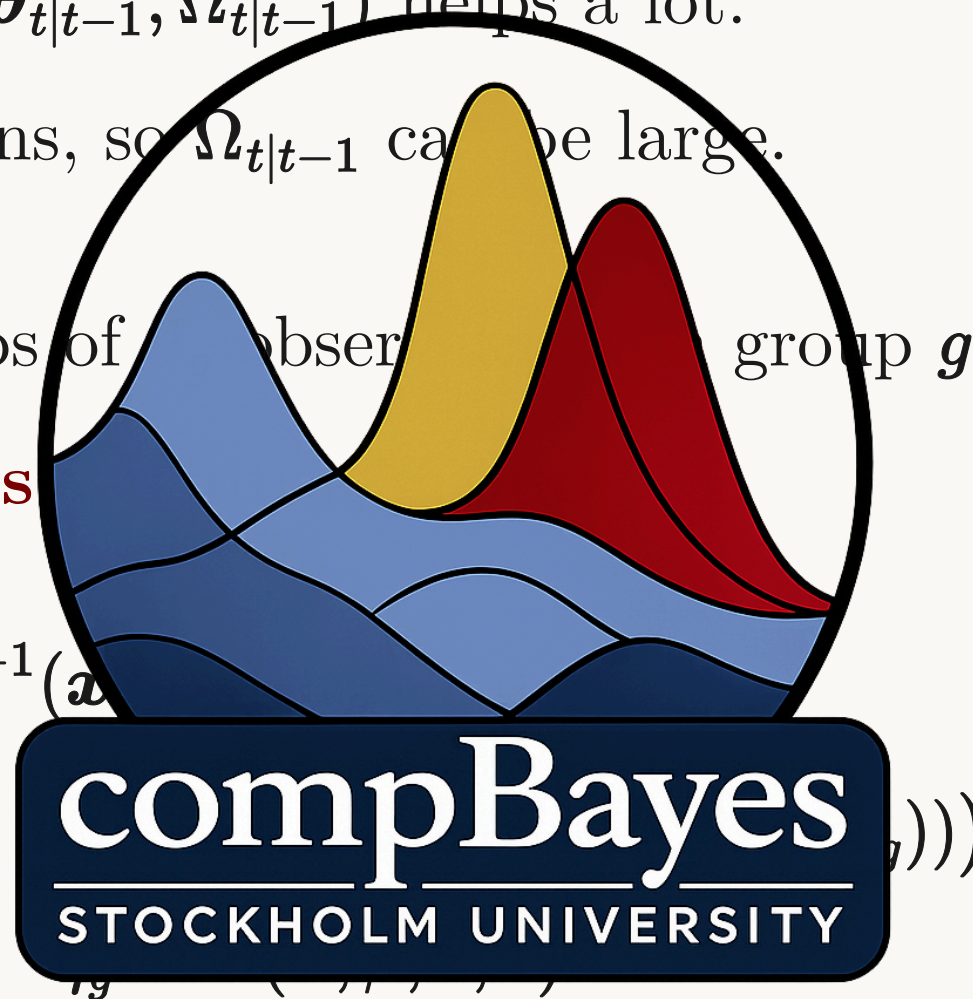
$$y_{gj}|\mathbf{x}_{gj} \sim \text{ExpFamily}\left(k^{-1}(\mathbf{x}_{gj}|\boldsymbol{\theta}_g)\right)$$

$$\boldsymbol{\beta}_g = \boldsymbol{\beta}_{g-1} + \boldsymbol{\nu}_g,$$

$$\mathbf{h}_g = \Phi \mathbf{h}_{g-1} + \boldsymbol{\eta}_g,$$

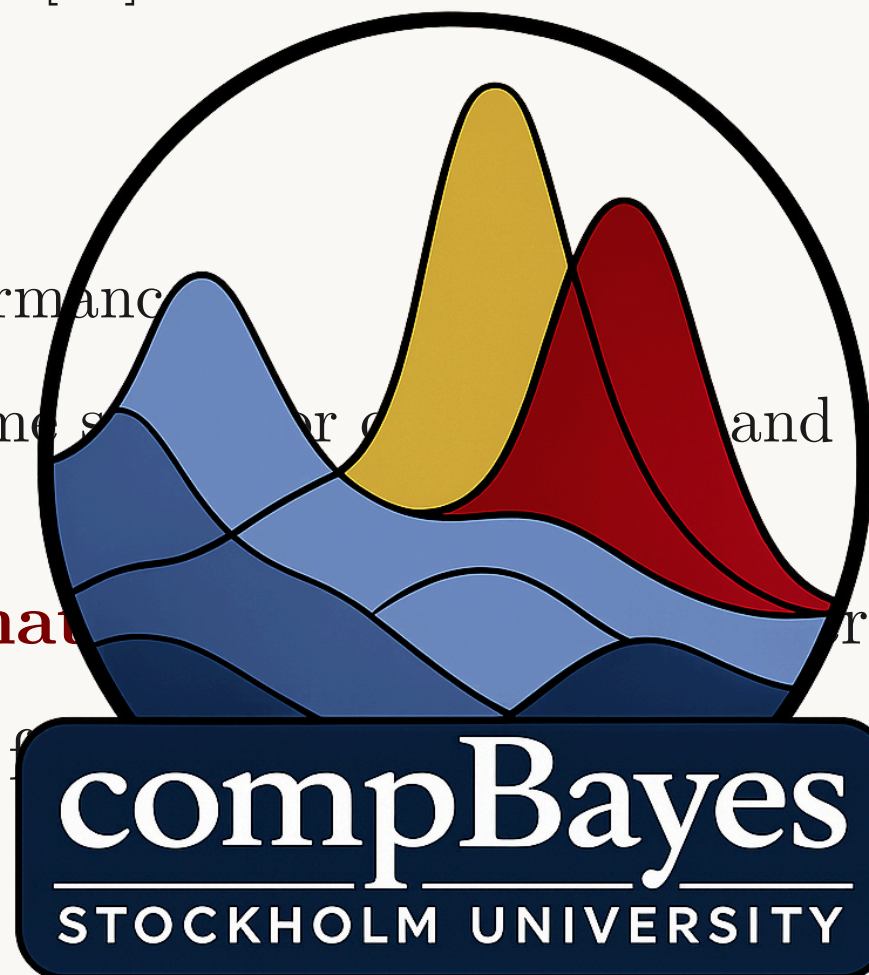
where $j = 1, \dots, J_g$ for $g = 1, \dots, G$.

“When all the hope is gone, there is no reason for pessimism”.
Aki Kaurismäki



Grouping observations

- Related ideas:
 - Improve Gaussianity in Subsampling MCMC/HMC [18]
 - Improve linearity/Gaussianity in filtering [19]
 - Modeling moderate time evolutions [20]
- Two views:
 - interpolation for algorithmic performance
 - genuine model change: different time series or different parameters.
- **Parameter evolution is automatic** for different group sizes.
- **Massive speed-up**, particularly for large groups.
- Example: 30 min electricity price. Observations by month.
- Large groups: Sequential filtering of individual observations within group.



Natural parameter innovations

- Parameter evolution should adapt to:
 - Scales of covariates
 - Link functions
 - Model geometry
- **Natural gradients** [21, 22] in optimization. Fisher metric.
- Parameter evolution with **natural innovation**

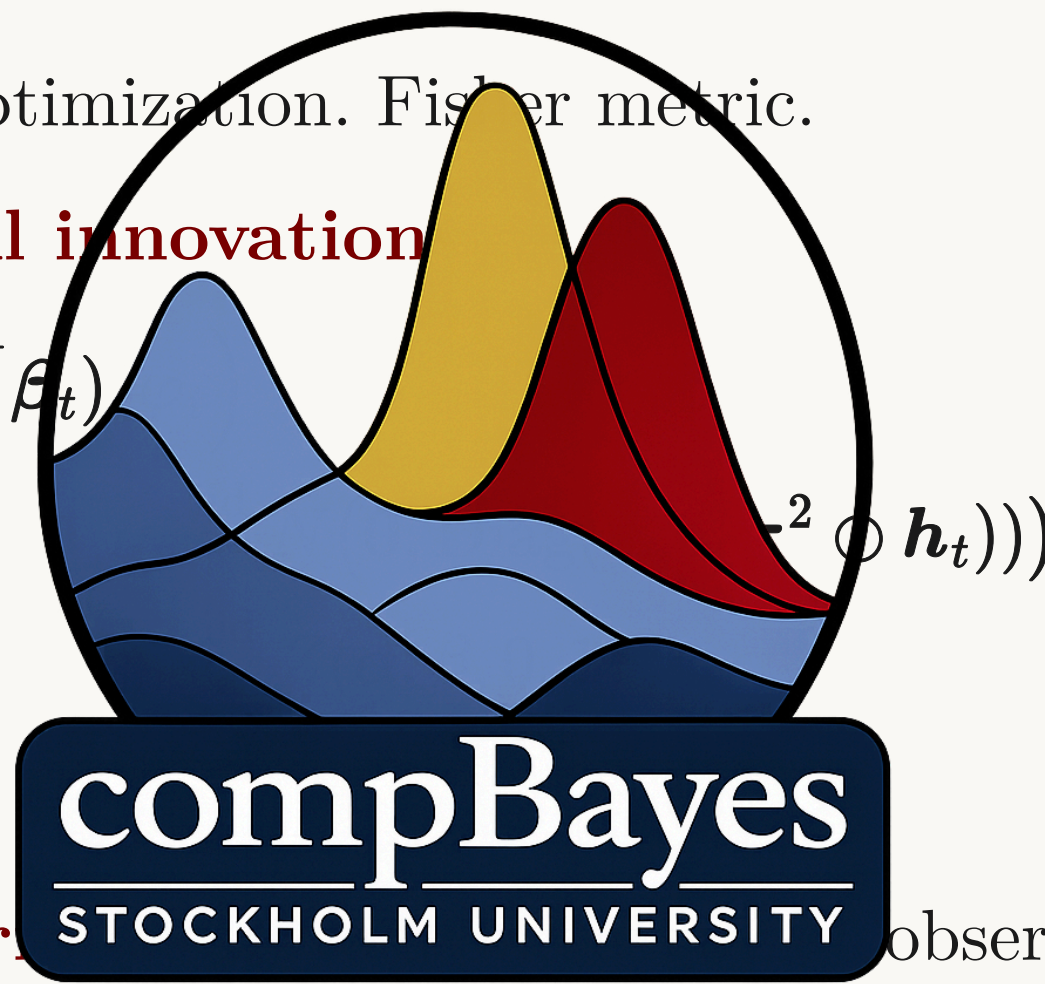
$$y_t | \mathbf{x}_t \sim \text{ExpFamily} \left(k^{-1}(\mathbf{x}_t^\top \boldsymbol{\beta}_t) \right)$$

$$\boldsymbol{\beta}_t = \boldsymbol{\beta}_{t-1} + \mathbf{S}(\boldsymbol{\beta}_{t-1}) \boldsymbol{\nu}_t,$$

$$\mathbf{h}_t = \Phi \mathbf{h}_{t-1} + \boldsymbol{\eta}_t,$$

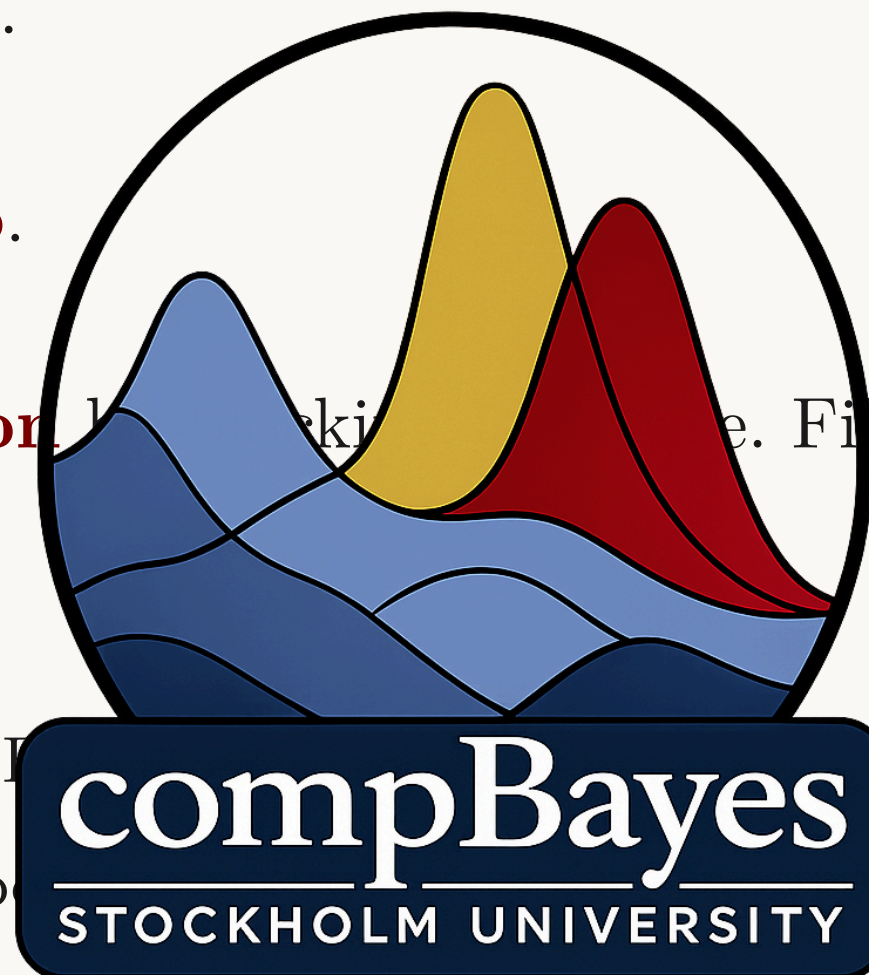
where $\mathbf{S}(\boldsymbol{\beta}_{t-1}) = \bar{\mathbf{I}}^{-1/2}(\boldsymbol{\beta}_{t-1})$

- $\bar{\mathbf{I}}(\boldsymbol{\beta}) = \frac{1}{T} \mathbf{I}(\boldsymbol{\beta})$ is the **Fisher information** observation.
- Non-linear transition model. Prior propagation step: $\mathbf{S}(\boldsymbol{\beta}_{t-1}) \approx \mathbf{S}(\hat{\boldsymbol{\beta}}_{t-1|t-1})$.



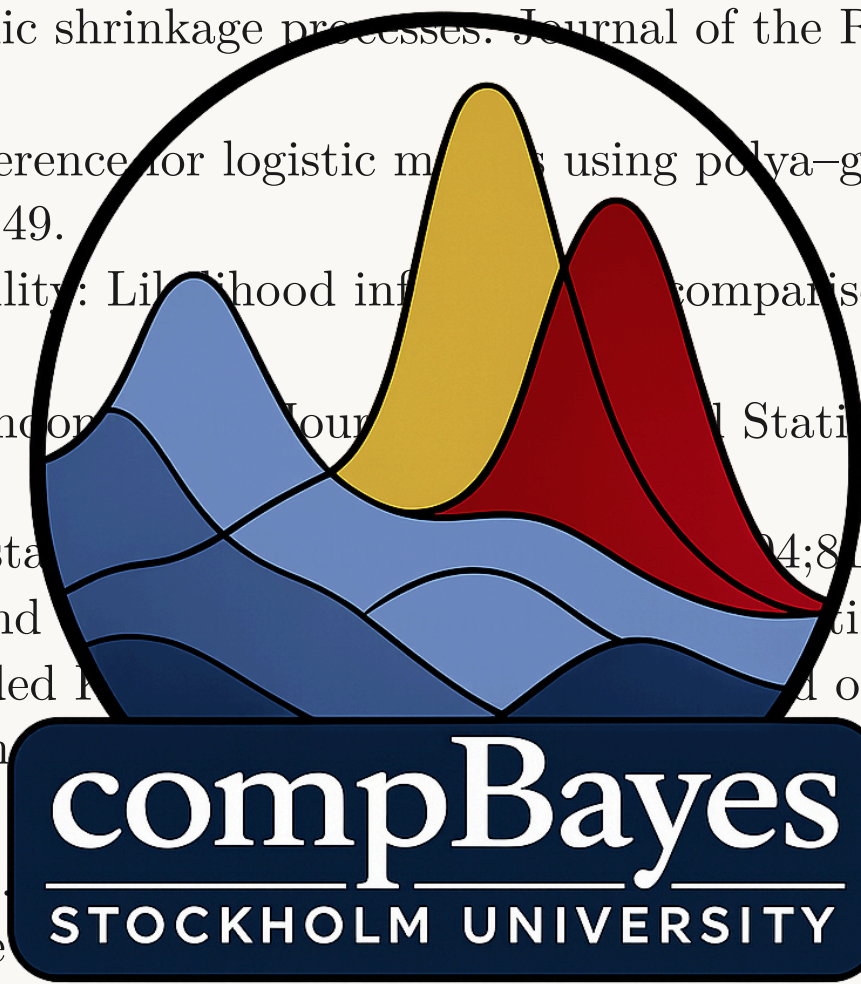
Ongoing and Future work

- **Multivariate** extensions.
- **Forecasting**.
- More **efficient** sampling of dsp.
- Better **prior elicitation** for dsp.
- **Metropolis-Hastings correction** for **ski**. Filtering version of [16].
- **Delayed rejection**:
 - first try with cheap proposal (e.g. **I**)
 - if rejected try more expensive proposal (e.g. **line search**)



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